AS · Unit 1 · WPH11 Mechanics & Materials

Motion

Motion is the change in position of an object over time. It can be described in terms of displacement, distance, speed, velocity, and acceleration. Understanding motion is fundamental to the study of physics as it helps explain how objects interact with forces and energy.

Key Concepts in Motion

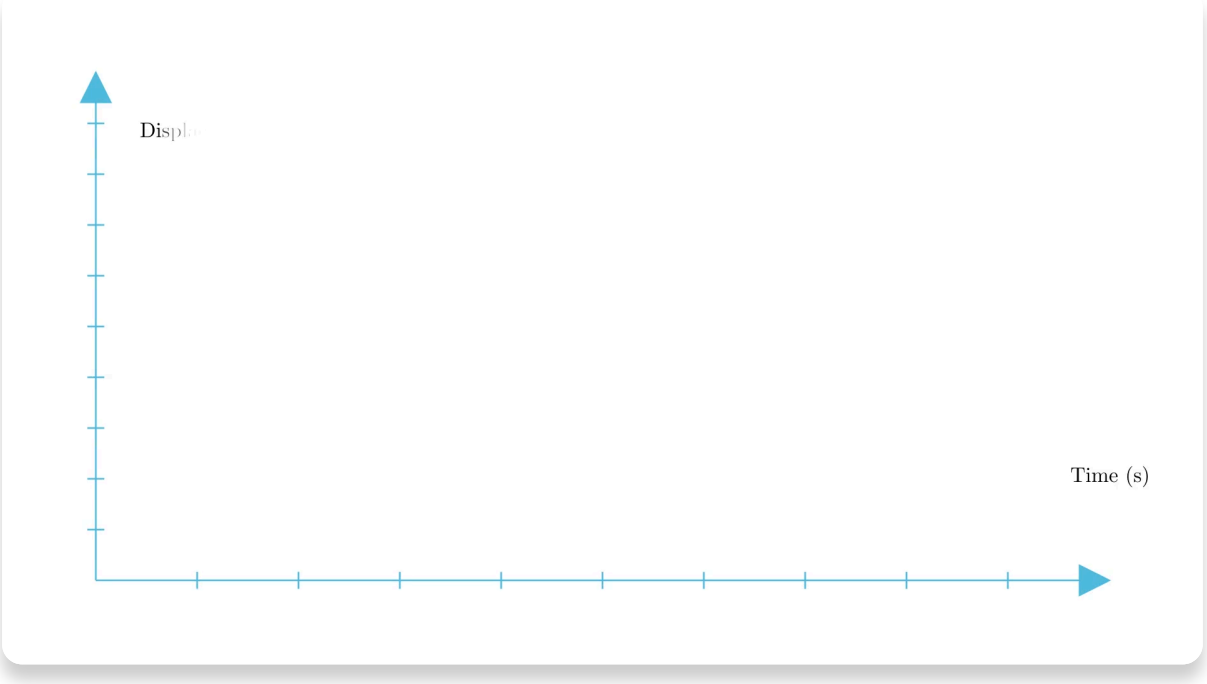
- **Displacement:** A vector quantity that refers to the change in position of an object. It has both magnitude and direction.
- **Distance:** A scalar quantity that refers to how much ground an object has covered during its motion.
- **Speed:** A scalar quantity that refers to how fast an object is moving. It is calculated as distance divided by time.
- **Velocity:** A vector quantity that refers to the rate at which an object changes its position. It is calculated as displacement divided by time ($v = \Delta s / \Delta t$).
- **Acceleration:** A vector quantity that refers to the rate at which an object's velocity changes over time. It is calculated as change in velocity divided by time ($a = \Delta v / \Delta t$).

Scalars (magnitude only)	Vectors (magnitude *and* direction)
distance, speed, time, mass	displacement, velocity, acceleration
energy, work, power, density	force, weight, momentum
temperature, charge, volume	electric field strength, gravitational field strength

EXAM TIP

Vectors are shown in print with bold letters (**F**) or with an arrow over them ($\vec{F}$). In a calculation you pick a positive direction first, then anything pointing the other way gets a minus sign. Most SUVAT mistakes come from forgetting this.

Interpreting displacement-time, velocity-time and acceleration-time graphs



As you can see, when displacement is increasing at a constant rate, where displacement is directly proportional to time, the velocity is constant.

This shows that obtaining the velocity graph from a displacement graph is just **differentiating** the function used to graph displacement: the velocity at any instant is the **gradient** of the s - t graph at that instant.

Just in case it is not clear yet, let's suppose this is the equation for displacement:

$$s = 10t$$

$$v = \frac{ds}{dt} = \frac{d}{dt}(10t)$$

$$\frac{d}{dt}(10t) = 10$$

Thus:

$$v = 10 \text{ m s}^{-1}$$

More s - t graph examples:

This example shows a parabolic s - t graph.

If we apply the same logic to this situation, we obtain a linear graph; this makes sense as our gradient (which represents the velocity at that point) is increasing at a linear rate, which also means our velocity is increasing linearly.

In this example, the gradient starts off steep, but slowly levels off to a gradient of 0.

Velocity time graphs

Velocity - time graphs have velocity on the **y - axis** and time on the **x - axis**. In this situation, the gradient of the graph at a point **represents the acceleration** of the object.

The **Area** under the graph until a certain point represents the **displacement** of the object (this is the same as the distance moved as long as the object never turns around; area below the time axis counts as negative displacement).

Here is an example:

Acceleration time graphs

For an **acceleration - time** graph, the area under the graph from the start until a certain time represents the **change in velocity** up to that time.

This means that when acceleration is **negative**, velocity is **not necessarily** negative, it is just decreasing.

Here is an example:

In this situation, the area under the **positive** part and **negative** part are **equal**.
Thus, if the object started from rest, the velocity at the end would be equal to **0 m/s**, as the areas (which represent the change in velocity) cancel out.

Try it yourself: all three graphs at once

Pick a motion, or choose **Drive it yourself** and push the acceleration slider around while the car moves. The dots on the road are where the car was every second, like a ticker-tape or multiflash photo, so wider spacing means faster. When it stops (or you press pause), move your mouse over any graph: you will see the **tangent** on the s - t graph (its gradient is v) and the **shaded area** under the v - t and a - t graphs (giving s and Δv).

KEY IDEA

Gradients and areas: the whole topic in one table

Graph	Gradient gives	Area under it gives
displacement–time	velocity	(nothing useful)
velocity–time	acceleration	displacement
acceleration–time	(rate of change of acceleration, not needed)	change in velocity

For a curved graph, the **instantaneous** rate of change is the gradient of the **tangent** at that point. The **average** rate of change is the gradient of the straight line joining two points.

Equations of Motion

The equations of motion describe the relationship between displacement, velocity, acceleration, and time for uniformly accelerated motion. The four main equations are:

FORMULA SHEET

$$v = u + at$$

FORMULA SHEET

$$s = \frac{1}{2}(u + v)t$$

FORMULA SHEET

$$s = ut + \frac{1}{2}at^2$$

FORMULA SHEET

$$v^2 = u^2 + 2as$$

Where:

- v = final velocity (m/s)
- u = initial velocity (m/s)
- a = acceleration (m/s²)
- s = displacement (m)
- t = time (s)

EXAM TIP

Each equation leaves out one of the five letters. Write down the three you know and the one you want; the equation that doesn't contain the fifth one is the one to use.

Derivation of Equations of Motion

They can be derived from the definitions of velocity and acceleration, along with the assumption of constant acceleration. For example, the first equation (v = u + at) comes directly from the definition of acceleration as the change in velocity over time. The others come from the area under the velocity - time graph.

Deriving all four SUVAT equations from a v–t graph

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 Acceleration is the rate of change of velocity. If it is constant, it is the change in velocity divided by the time taken:

$$a = \frac{v - u}{t} \quad \Rightarrow \quad \boxed{v = u + at}$$

This is also why the v–t graph is a straight line: the gradient is constant.

2 Displacement is the area under the v–t graph. The graph is a straight line from u up to v, so the area is a trapezium of width t:

$$\boxed{s = \tfrac{1}{2}(u + v)\,t}$$

3 Now split that trapezium into a rectangle (height u) and a triangle on top (height $v - u$):

$$s = ut + \tfrac{1}{2}\,t\,(v - u)$$

4 From step 1, $v - u = at$. Substitute it in:

$$s = ut + \tfrac{1}{2}\,t\,(at) \quad \Rightarrow \quad \boxed{s = ut + \tfrac{1}{2}at^2}$$

5 To get an equation without t, rearrange step 1 to $t = \dfrac{v - u}{a}$ and put it into the trapezium equation:

$$s = \tfrac{1}{2}(u + v) \cdot \frac{(v - u)}{a} = \frac{v^2 - u^2}{2a}$$

$(u + v)(v - u)$ *is a difference of two squares:* $v^2 - u^2$.

6 Multiply both sides by $2a$ and rearrange:

$$\boxed{v^2 = u^2 + 2as}$$

Further Insight

As you saw in the animation, these equations all assume that acceleration is constant. The point of showing the derivation of these equations is to give a deeper understanding and insight into this topic, as that is vital to being able to solve the questions given in the exams. Physics is not about memorizing, but about understanding.

Another, more mathematical, way to derive these equations is to use integration:

SI Units

The last thing you need to know are the S.I. base units. These are base units which are what every other form of measurement are compared to. They serve as the modern form of the metric system, used world wide. They are:

- Meter (m): for length
- Kilogram (kg): for mass
- Second (s): for time
- Ampere (A): for electric current
- Kelvin (K): for temperature
- Mole (mol): for amount of substance
- Candela (cd): for light intensity (you won't need this one for Edexcel)

Here are some examples of derived units:

- Newton (N): the unit of force, equivalent to $kg \cdot m/s^2$
- Joule (J): the unit of work or energy, equivalent to $N \cdot m$
- Pascal (Pa): the unit of pressure, equivalent to N/m^2

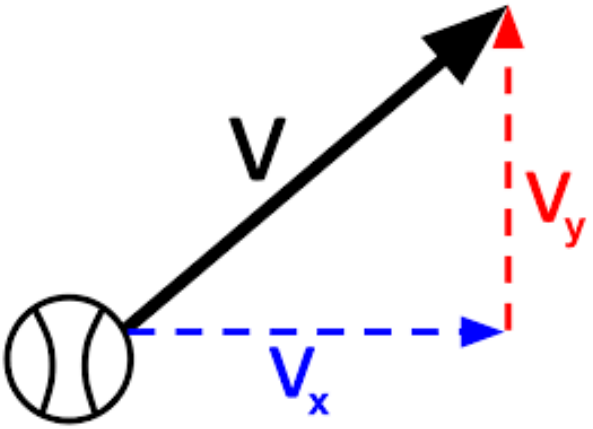
- Watt (W): the unit of power, equivalent to J/s
- Meters per second (m/s): the unit for velocity and speed
- Meters per second squared (m/s^2): the unit for acceleration
- Cubic meters (m^3): the unit for volume
- Square meters (m^2): the unit for area

EXAM TIP

To find the base units of anything, write it as an equation of things you already know. For example $W = Fs$, so $1\ \mathrm{J} = 1\ \mathrm{N} \times 1\ \mathrm{m} = 1\ \mathrm{kg\ m^2\ s^{-2}}$. An equation must be **homogeneous**: both sides have the same base units.

Vectors: resolving and adding

To understand projectile motion, you first need to understand the nature of vector quantities and how they can be broken up into vertical and horizontal components.



As a ball is moving at an angle, it is moving forward a certain velocity and upwards a certain velocity. We can calculate these velocity components using our SohCahToa rules as V_y and V_x are perpendicular.
Thus, we can write V_y and V_x as:

$$V_y = V \cdot \sin(\theta)$$

$$V_x = V \cdot \cos(\theta)$$

where θ is the **angle** the ball is thrown at, measured from the horizontal.

KEY IDEA

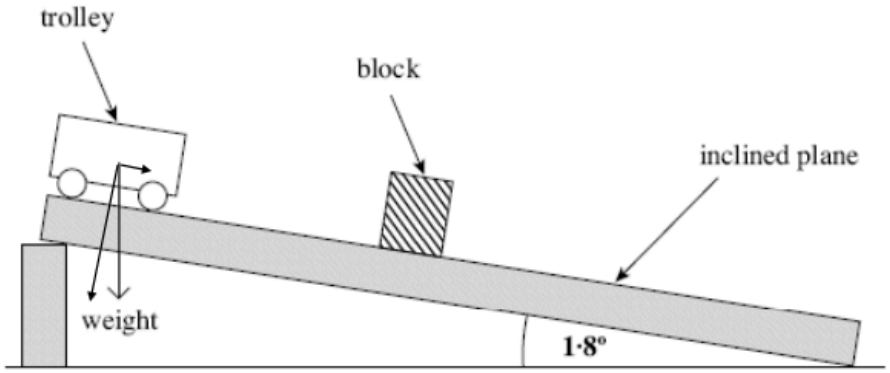
Adding two vectors

Draw them **tip-to-tail**; the resultant joins the start of the first to the end of the second. If the two vectors are at right angles you can calculate the resultant with Pythagoras and its direction with tan:

$$R = \sqrt{A^2 + B^2}, \qquad \tan\theta = \frac{B}{A}$$

For any other angle, the spec only expects you to find the resultant with a **scale drawing** (or by resolving each vector into components and adding those).

Now let's do a question (**remember**, this concept applies to all vector quantities, such as force, momentum etc.)



The mass of the trolley in the figure above is 0.20 kg and the slope is at an angle of 1·8° to the horizontal.

(i) Show that the component of the weight acting along the slope is about 0.06 N.

First, let us find the weight of the trolley...

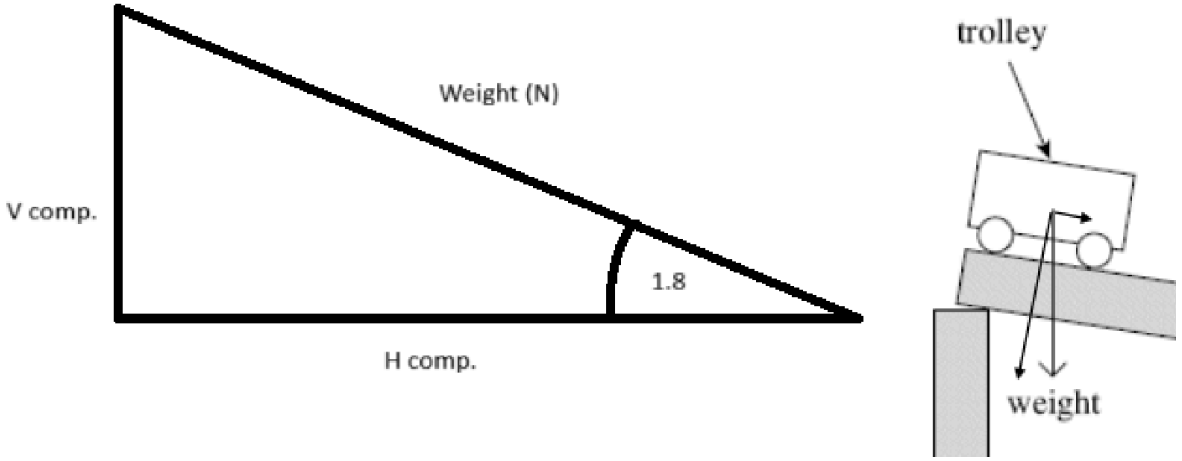
we know that:

$$W = m \cdot g$$

Thus

$$W = 0.2 \cdot 9.81 = 1.962 \text{ N}$$

Now let us use the formula from before (remember, we are trying to find the component of the weight along the slope, which is the vertical component in the diagram):



$$V_y = V \cdot \sin(\theta), \qquad \theta = 1.8^\circ$$

$$V_y = 1.962 \cdot \sin(1.8^\circ) = 0.0616 \text{ N } (3 \text{ s.f.}) \approx 0.06 \text{ N}$$

COMMON MISTAKE

On a slope of angle θ, the component of the weight **along** the slope is $mg\sin\theta$ and the component **perpendicular** to it is $mg\cos\theta$. Check with an extreme case: when $\theta = 0$ (flat ground) nothing pulls the trolley along, and $\sin 0 = 0$.

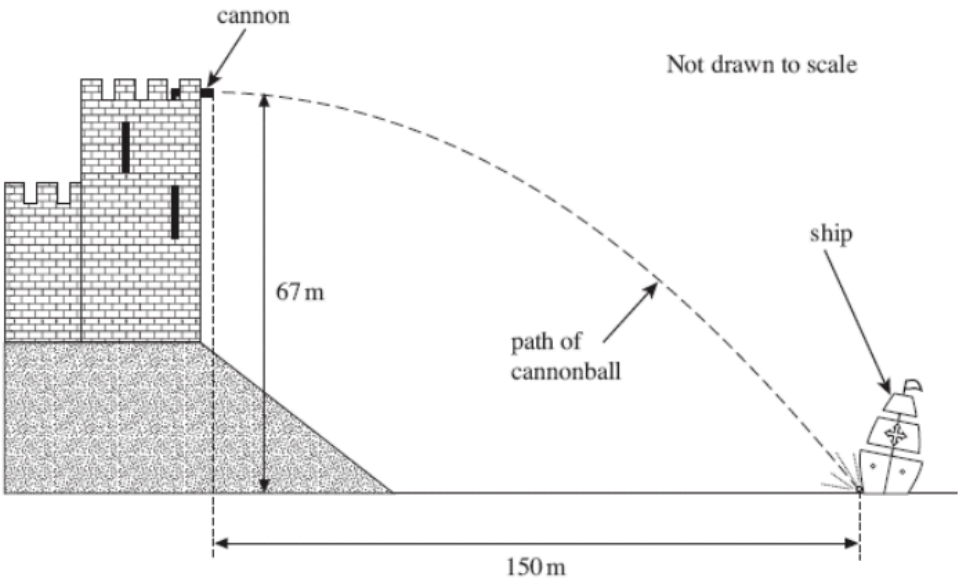
Projectile motion

A projectile is anything moving freely under gravity alone. The big idea is that the **horizontal and vertical motions are independent**: gravity only acts vertically, so

- **Horizontally:** no force (ignoring air resistance), so $a_x = 0$ and the horizontal velocity $u\cos\theta$ never changes.
- **Vertically:** constant acceleration $g = 9.81 \text{ m s}^{-2}$ downwards, so use SUVAT with $u_y = u\sin\theta$.
- **Time** is the only thing the two directions share. Find it from one direction and use it in the other.

Now let's look at a projectile motion question:

Q6. In a castle, overlooking a river, a cannon was once employed to fire at enemy ships. One ship was hit by a cannonball at a horizontal distance of 150 m from the cannon as shown in the figure below. The height of the cannon above the river was 67 m and the cannonball was fired horizontally.



(a) (i) Show that the time taken for the cannonball to reach the water surface after being fired from the cannon was 3.7 s. Assume the air resistance was negligible.

In this question, we are given the height the ball was thrown from, the distance between the cannon and the ship, and the initial vertical velocity (which is 0 as it is shot horizontally).

Out of the 2 distances given, the one we will use is the vertical one. We choose this one as acceleration due to gravity (9.81 m/s²) acts downwards, which causes the ball to fall. Now, we can use the SUVAT equations we derived previously (to choose which equation to use, compare the values you are given and what you have to find with the equations provided on the sheet, then choose accordingly).

Taking **down as positive**:

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$s_y = 67, \quad u_y = 0, \quad a_y = 9.81$$

$$67 = 0 \cdot t + \frac{1}{2} \cdot 9.81\, t^2$$

Rearrange:

$$t^2 = \frac{2 \cdot 67}{9.81} = 13.66 \quad \Rightarrow \quad t = 3.696 \text{ s} \approx 3.7 \text{ s}$$

(ii) Calculate the velocity at which the cannonball was fired. Give your answer to an appropriate number of significant figures.

For this question, we need to use the **t** value we calculated previously. As we know the distance to the ship and the time it takes to reach it, we can use the v = s/t equation to work out its initial horizontal velocity (which stays constant):

$$V_x = \frac{S_x}{t} = \frac{150}{3.696} = 40.6 \text{ m s}^{-1} \text{ (3 s.f.)}$$

(iii) Calculate the vertical component of velocity just before the cannonball hit the ship.

Now, we can use the value of t and the acceleration equation to find out the vertical component of velocity just before the ball hits the ship (which is at t = 3.696 s):

$$a_y = \frac{v_y - u_y}{t} \quad \Rightarrow \quad v_y = a_y \cdot t + u_y$$

$$v_y = 9.81 \cdot 3.696 + 0 = 36.3 \text{ m s}^{-1} \text{ (3 s.f.)}$$

(iv) By calculation or scale drawing, find the magnitude and direction of the velocity of the cannonball just before it hit the ship.

To do this question, we use our answers from parts (ii) and (iii) and apply our knowledge of vector components.

First we need to find the magnitude of the velocity, which is simple as we use Pythagoras' theorem. As the question explicitly states **just before it hits the ship**, our v_y and v_x are 36.3 and 40.6:

$$V = \sqrt{v_x^2 + v_y^2} = \sqrt{40.6^2 + 36.3^2} = 54.5 \text{ m s}^{-1} \text{ (3 s.f.)}$$

Now, we need to find the direction, which is the angle at which it is travelling during that time. To calculate this, we will use SohCahToa:

$$\tan(\theta) = \frac{OPP}{ADJ} = \frac{v_y}{v_x} = \frac{36.3}{40.6}$$

$$\theta = \arctan\left(\frac{36.3}{40.6}\right) = 41.8^\circ \text{ below the horizontal}$$

As you can see, projectile motion is nothing more than applying your knowledge of SUVAT equations and vectors, there is nothing *new*.

Core Practical 1: measuring g

CORE PRACTICAL

Determine the acceleration of a freely-falling object

A steel ball is held by an electromagnet and released; a timer starts when the current is switched off and stops when the ball breaks a light gate (or opens a trapdoor) a height h below. Repeat for several heights.

Because the ball starts from rest, $h = \frac{1}{2}gt^2$. This has the form $y = mx$, so plotting h **against** t^2 gives a straight line through the origin with **gradient** $= g/2$.

- Measure h from the **bottom** of the ball to the light gate, and use a set square to avoid parallax.
- Repeat each height and average to reduce the effect of random error.
- A delay in the magnet releasing the ball is a **systematic** error: every time is too long, so the line no longer goes through the origin. Using the gradient (not a single reading) removes its effect on g.

Quick checks

QUICK CHECK

A ball is thrown vertically upwards. At the very top of its flight, which statement is correct?

- Its velocity is zero and its acceleration is zero.
- Its velocity is zero and its acceleration is 9.81 m s^{-2} downwards.
- Its velocity is at a maximum and its acceleration is zero.
- Its velocity is zero and its acceleration is upwards.

Gravity never switches off, so the acceleration is g downwards for the whole flight. At the top the velocity is changing from positive (up) to negative (down), so it passes through zero.

QUICK CHECK

What does the area under an acceleration–time graph represent?

- Displacement
- Change in velocity
- Distance travelled
- Final velocity, always

Area = $a \times t = \Delta v$. It only equals the final velocity if the object started from rest.

QUICK CHECK

One ball is dropped from a cliff at the same moment another is fired horizontally at 20 m s^{-1} from the same height. Ignoring air resistance, which lands first?

- The dropped ball
- The fired ball
- They land at the same time
- It depends on their masses

Both have $u_y = 0$ and $a_y = g$, so the vertical motion is identical. The horizontal velocity of the fired ball has no effect on how fast it falls.

QUICK CHECK

A force of 10 N acts at 30° above the horizontal. What is its horizontal component?

- 5.0 N
- 8.7 N
- 10 N
- 11.5 N

The horizontal component is adjacent to the angle: $10\cos 30^\circ = 8.66\ \text{N}$. A component can never be larger than the vector itself.

AS · Unit 1 · WPH11 Mechanics & Materials

Forces & Momentum

Motion tells you *how* things move. Forces tell you *why*. On this page you'll learn to draw free-body diagrams, use Newton's three laws, see why a skydiver stops speeding up, and use momentum to predict what happens when things collide or fly apart. Nearly every mechanics question starts with a free-body diagram, so that's where we begin.

Forces and free-body diagrams

A **force** is a push or a pull. It is a **vector**, so it has a size (measured in **newtons**, N) and a direction. Forces come from interactions between objects, and in mechanics you will keep meeting the same handful:

Force	Where it comes from	Direction
Weight $W = mg$	the Earth's gravitational pull on the object	vertically down, acting at the centre of gravity
Normal reaction R	a surface pushing back on an object resting on it	perpendicular to the surface, away from it
Friction	two surfaces rubbing or trying to slide over each other	along the surface, opposing the (attempted) sliding
Tension T	a stretched string, rope or cable	along the string, pulling away from the object
Drag (air resistance)	moving through a fluid; it grows with speed	opposite to the velocity
Upthrust	a fluid pushing up on anything in it (see Density, Upthrust & Viscous Drag)	vertically up

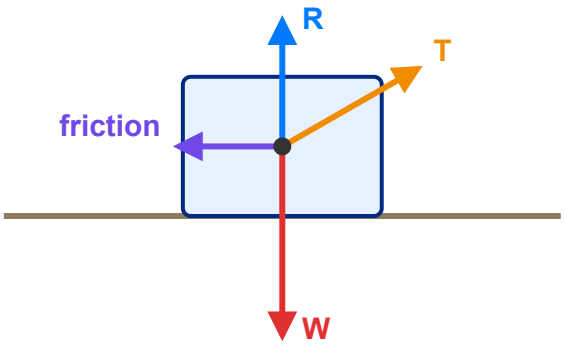
A **free-body diagram** shows **one** object on its own, with an arrow for every force acting **on** it. The rules:

- Draw only the body you are interested in, not the things around it.
- Include only forces acting **on** that body, never forces it exerts on other things.
- Each arrow points the way the force acts, with length roughly showing its size. Label every arrow.
- Don't draw the resultant force or "the force of motion". There is no such force; the resultant is what you *calculate* from the diagram.

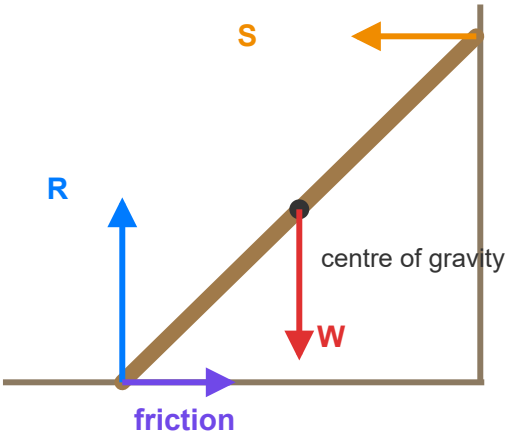
Particles and extended bodies

If you only care about how an object moves as a whole (not whether it spins), you can model it as a **particle** and draw every force from one point. If the object could rotate, like a ladder or a beam, it is an **extended body**: each force must be drawn where it actually acts, and the weight acts at the **centre of gravity**. Where a force acts matters for turning, which is the whole of the Moments page.

Particle model
box dragged by a rope



Extended body
ladder leaning on a smooth wall



Left: the box won't rotate, so every force can be drawn from one point. Right: the ladder can rotate, so each force is drawn where it acts, with the weight at the centre of gravity. The smooth wall can only push at right angles to itself, so friction at the foot stops the ladder sliding out.

DEFINITION

Centre of gravity

The **centre of gravity** of an object is the single point through which its whole weight can be taken to act. For a uniform object it is at the geometric centre.

Forces on a slope

On a slope, the trick is to **resolve along and perpendicular to the slope**, because that is the direction the object can move. The weight splits into two components (exactly as on the Motion page's vectors section):

$$\text{down the slope: } mg\sin\theta \qquad \text{into the slope: } mg\cos\theta$$

Nothing accelerates into or out of the surface, so perpendicular to the slope the forces balance and $R = mg\cos\theta$ (as long as no other force has a perpendicular component). Along the slope, whatever is left over is the resultant force that accelerates the object.

EXAM TIP

Check your components with an extreme case. At $\theta = 0$ (flat ground) nothing pulls the block along, and $\sin 0 = 0$. At $\theta = 90^\circ$ (a vertical wall) the whole weight acts along the "slope", and $\sin 90^\circ = 1$. If your expression doesn't behave like that, you've swapped sin and cos.

Newton's first and second laws

DEFINITION

Newton's first law

An object remains at rest, or continues to move with constant velocity, unless a resultant force acts on it.

This sounds obvious until you notice what it rules out. You do **not** need a force to keep something moving; you need a force to **change** its velocity. A hockey puck on smooth ice glides on and on. A car on the motorway at a steady 30 m s^{-1} still needs its engine only because the driving force has to cancel air resistance and friction. When all the forces on an object cancel, we say it is in **equilibrium**, and it is either at rest or moving at constant velocity. Those two cases look different to us, but to Newton they are the same.

When the forces don't cancel, the **resultant force** makes the object accelerate. Experiments show the acceleration is proportional to the resultant force (push twice as hard, get twice the acceleration) and inversely proportional to the mass (twice the mass, half the acceleration). Mass measures **inertia**: how strongly an object resists a change in its motion.

Newton's second law

FORMULA SHEET

$$F = ma$$

F: resultant force / N m: mass / kg a: acceleration / m s^{-2}

The newton is defined by this equation: **1 N is the resultant force that gives a mass of 1 kg an acceleration of 1 m s^{-2}**, so $1\ \text{N} = 1\ \text{kg m s}^{-2}$. The acceleration is always in the same direction as the resultant force. Newton's first law is just the special case $a = 0$: no resultant force, no acceleration.

COMMON MISTAKE

F in $F = ma$ is the **resultant** force, not "the force" or the biggest force. Choose a positive direction, add up the forces that way, and subtract the ones the other way. This form of the law also only works when the mass stays constant. (Rockets, which throw mass away, need the momentum form further down.)

WORKED EXAMPLE

A car of mass 1200 kg has a driving force of 3000 N. The total resistive force on it is 900 N. (a) Calculate its acceleration. (b) Assuming the forces stay constant, how long does it take to reach 20 m s^{-1} from rest?

Show solution

(a) Taking the direction of motion as positive, the resultant force is

$$F = 3000 - 900 = 2100 \text{ N} \qquad a = \frac{F}{m} = \frac{2100}{1200} = 1.75 \text{ m s}^{-2}$$

(b) The acceleration is constant, so use $v = u + at$:

$$t = \frac{v - u}{a} = \frac{20 - 0}{1.75} = 11.4 \text{ s}$$

WORKED EXAMPLE

A car of mass 1000 kg tows a trailer of mass 500 kg along a level road. The driving force is 2400 N. The resistive forces are 300 N on the car and 200 N on the trailer. Calculate (a) the acceleration and (b) the tension in the tow bar.

Show solution

(a) Treat the car and trailer as one body of mass 1500 kg. The tension is internal (the car pulls the trailer, the trailer pulls back on the car), so it cancels out:

$$a = \frac{2400 - 300 - 200}{1500} = \frac{1900}{1500} = 1.27 \text{ m s}^{-2}$$

(b) Now draw a free-body diagram of the trailer alone. Forwards: tension T. Backwards: 200 N. Apply $F = ma$ to the trailer:

$$T - 200 = 500 \times 1.267 \quad \Rightarrow \quad T = 833 \text{ N}$$

Check with the car: $2400 - 300 - 833 = 1267 \text{ N} = 1000 \times 1.267$. It works.

Lifts and bathroom scales

Bathroom scales don't measure your weight. They measure how hard they have to push up on you: the normal reaction R. Taking **up as positive**, Newton's second law for you in a lift gives

$$R - mg = ma \quad \Rightarrow \quad R = m(g + a)$$

If the lift accelerates upwards (a positive) the scales must push harder than your weight, and you feel heavier. If it accelerates downwards they push less, and you feel lighter. At constant velocity, in either direction, $a = 0$ and the scales read exactly mg. In free fall $a = -g$, so $R = 0$: you feel weightless even though gravity is still pulling on you as strongly as ever.

WORKED EXAMPLE

A 65 kg student stands on scales in a lift. Find the reading of the scales, in newtons, when the lift (a) accelerates upwards at 1.2 m s^{-2}, (b) moves up at a constant 3.0 m s^{-1}, (c) slows down at 1.2 m s^{-2} while moving up.

Show solution

Up is positive, so $R = m(g + a)$.

$$\text{(a)} \quad R = 65(9.81 + 1.2) = 716 \text{ N} \qquad \text{(b)} \quad R = 65 \times 9.81 = 638 \text{ N}$$

(c) Slowing down while moving up means the acceleration is **downwards**, so $a = -1.2 \text{ m s}^{-2}$:

$$R = 65(9.81 - 1.2) = 560 \text{ N}$$

Weight and gravitational field strength

Mass is the amount of matter in an object and its resistance to acceleration. It is a scalar, measured in kg, and it is the same everywhere. **Weight** is the gravitational force on the object. It is a vector, measured in N, and it depends on where you are. The link between them is the **gravitational field strength** g: the force per unit mass on an object in the field.

Gravitational field stre

FORMULA SHEET

$$g = \frac{F}{m}$$

g: field strength / N kg^{-1}

F: gravitational force / N

Weight

FORMULA SHEET

$$W = mg$$

W: weight / N m: mass / kg

DEFINITION

Gravitational field strength

The gravitational field strength at a point is the gravitational force per unit mass acting on a small mass placed at that point. Near the Earth's surface $g = 9.81\ \text{N kg}^{-1}$.

Now put $W = mg$ into Newton's second law for an object falling with no air resistance. The only force is its weight, so $a = \dfrac{F}{m} = \dfrac{mg}{m} = g$. That's why every object falls with the same acceleration: a heavier object is pulled harder, but it also has more inertia, and the two effects cancel exactly. It also shows that $9.81\ \text{N kg}^{-1}$ and $9.81\ \text{m s}^{-2}$ are the same thing. To measure g yourself, see Core Practical 1 on the Motion page.

WORKED EXAMPLE

An astronaut has a mass of 80 kg. On the Moon $g = 1.62\ \text{N kg}^{-1}$. (a) Find her weight on Earth and on the Moon. (b) What horizontal resultant force would give her an acceleration of 2.0 m s⁻² on the Moon?

Show solution

$$\text{(a)}\quad W_{\text{Earth}} = 80 \times 9.81 = 785\ \text{N}, \qquad W_{\text{Moon}} = 80 \times 1.62 = 130\ \text{N}$$

(b) $F = ma = 80 \times 2.0 = 160\ \text{N}$, exactly the same as on Earth. Her **weight** changed, but her **mass** (her inertia) did not. A Moon walker finds things easy to lift but just as hard to push into motion sideways.

Terminal velocity

Drop something through air or water and the drag on it grows as it speeds up. Follow a skydiver through Newton's second law, taking down as positive:

1. **Just after jumping**, $v = 0$ so there is no drag. The resultant force is the weight and $a = g$.
2. **As she speeds up**, the drag D increases, so the resultant force $W - D$ gets smaller. She is still speeding up, but the acceleration decreases: the v–t graph curves over.
3. **Eventually** the drag equals her weight. The resultant force is zero, so (Newton's first law) she carries on at a constant velocity: her **terminal velocity**.
4. **Opening the parachute** makes the drag at that speed far bigger than her weight. The resultant force is now upwards, so she **decelerates**. As she slows the drag falls, until it equals her weight again at a new, much lower terminal velocity.

DEFINITION

Terminal velocity

The constant, maximum velocity reached by an object falling through a fluid, when the drag (together with any upthrust) balances its weight, so the resultant force is zero.

EXAM TIP

"Describe and explain the motion" questions want the chain of reasoning, not just "it reaches terminal velocity". Say what happens to the drag, then the resultant force, then the acceleration, then the gradient of the v–t graph. A small ball falling through oil in Core Practical 2 does exactly the same thing, with its drag given by Stokes' law (see Density, Upthrust & Viscous Drag).

WORKED EXAMPLE

A skydiver and her kit have a total mass of 85 kg. (a) What is the drag on her at terminal velocity? (b) Just after her parachute opens, the drag is 2400 N. Calculate her acceleration at that moment.

Show solution

(a) At terminal velocity the resultant force is zero, so drag = weight:

$$D = mg = 85 \times 9.81 = 834 \text{ N}$$

(b) The resultant force is now upwards:

$$F = 2400 - 834 = 1566 \text{ N} \qquad a = \frac{1566}{85} = 18.4 \text{ m s}^{-2} \text{ upwards}$$

She is still moving **down**; the upward acceleration means she is slowing down.

Newton's third law

DEFINITION

Newton's third law

If body A exerts a force on body B, then body B exerts a force on body A that is equal in magnitude and opposite in direction.

Forces always come in pairs because they come from **interactions**. You can't push without being pushed back: lean on a wall and the wall leans on you. The two forces in a Newton's third law pair always:

- act on **two different bodies** (one on each);
- are the **same type** of force (both gravitational, both contact, both magnetic…);
- are **equal in magnitude** and **opposite in direction**, along the same line;
- exist for exactly the **same time**: they start and stop together.

Examples are everywhere. A swimmer pushes water backwards and the water pushes her forwards. A car's tyres push the road backwards and the road pushes the car forwards. The Earth pulls the Moon, and the Moon pulls the Earth just as hard (that pull on the Earth is what raises the tides).

COMMON MISTAKE

The classic wrong answer: "the weight of a book on a table and the normal reaction form a Newton's third law pair". They don't. Both act on the **same** body (the book) and they are different **types** (gravitational and contact). The partner of the book's weight is the book's gravitational pull **on the Earth**. The partner of the table's push on the book is the book's push **on the table**.

Momentum

Which is harder to stop: a tennis ball at 50 m s⁻¹ or a lorry at 1 m s⁻¹? Both mass and velocity matter, and the quantity that combines them is **momentum**.

Momentum

FORMULA SHEET

$$p = mv$$

p: momentum / kg m s⁻¹ m: mass / kg v: velocity / m s⁻¹

Momentum is a **vector**, in the same direction as the velocity, so in one-dimensional problems it carries a sign. A 2 kg trolley moving left at 3 m s⁻¹ has momentum $-6\ \text{kg m s}^{-1}$ if you chose right as positive. Its unit can also be written N s, since $1\ \text{N s} = 1\ \text{kg m s}^{-2} \times \text{s} = 1\ \text{kg m s}^{-1}$.

PREVIEW · UNIT 4

Force is the rate of change of momentum

You'll meet this properly in Unit 4 (on the Impulse & 2D Collisions page), but it's worth seeing now how Newton's second law connects to momentum.

From F = ma to F = Δp/Δt

1. Start from Newton's second law and write the acceleration as change in velocity over time:

$$F = ma = m\,\frac{v - u}{\Delta t}$$

2. If the mass is constant it can go inside the bracket:

$$F = \frac{mv - mu}{\Delta t} = \frac{\Delta p}{\Delta t}$$

*The resultant force equals the rate of change of momentum. Multiplying by Δt gives the **impulse** $F\Delta t = \Delta p$, which is why airbags and crumple zones work: making Δt longer makes the force for the same Δp smaller.*

Conservation of linear momentum

DEFINITION

Principle of conservation of linear momentum

The total linear momentum of a system of interacting bodies remains constant, provided no resultant external force acts on the system.

"External" is the important word. In a collision the bodies push on each other very hard, but those are internal forces. Newton's third law guarantees they come in equal and opposite pairs, so whatever momentum one body gains, the other loses. Here's the argument in steps.

Conservation of momentum from Newton's second and third laws

1. Two bodies A and B collide. During the collision, A pushes on B with force F. By Newton's third law, B pushes on A with force $-F$, and the two forces last for the same time Δt.

2. Newton's second law in momentum form, $F = \Delta p/\Delta t$, applied to each body:

$$\Delta p_B = F\,\Delta t \qquad\qquad \Delta p_A = -F\,\Delta t$$

3. Add them to get the change in the total momentum:

$$\Delta p_A + \Delta p_B = -F\Delta t + F\Delta t = 0$$

The total cannot change. If an outside force (friction, say) also acted, it would add its own $F\Delta t$ and the total would change, which is why the principle needs "no resultant external force".

4. So for any collision or explosion in one dimension:

$$\boxed{m_A u_A + m_B u_B = m_A v_A + m_B v_B}$$

Collisions and explosions

- **Collisions:** write the total momentum before and after, with signs, and solve for the unknown. If the bodies stick together, they share one final velocity.
- **Explosions and recoil:** if everything starts at rest, the total momentum is zero before, so it is zero after. The pieces fly apart with equal and opposite momenta: $m_A v_A = -m_B v_B$. The lighter piece goes faster. A gun recoils, a skater pushing off a wall glides backwards.

EXAM TIP

Momentum is conserved in every collision and explosion (with no external forces). **Kinetic energy usually isn't**: in a collision some is transferred to thermal energy and sound, and in an explosion extra kinetic energy comes from chemical or elastic stores. Never use "KE before = KE after" unless you are told the collision is elastic. Checking whether a collision is elastic is Unit 4 content.

WORKED EXAMPLE

A trolley of mass 0.80 kg moving at 0.50 m s^{-1} hits a stationary trolley of mass 1.20 kg. They stick together. (a) Find their velocity after the collision. (b) How much kinetic energy is transferred to other stores?

Show solution

(a) $0.80 \times 0.50 + 1.20 \times 0 = (0.80 + 1.20)\,v \quad \Rightarrow \quad v = \dfrac{0.40}{2.00} = 0.20 \text{ m s}^{-1}$

(b) $E_{k,\text{before}} = \frac{1}{2}(0.80)(0.50)^2 = 0.100 \text{ J}, \qquad E_{k,\text{after}} = \frac{1}{2}(2.00)(0.20)^2 = 0.040 \text{ J}$

So 0.060 J (60% of the kinetic energy) is transferred, mostly to thermal energy and sound.

WORKED EXAMPLE

A rifle of mass 4.0 kg fires a bullet of mass 10 g at 400 m s⁻¹. Calculate the recoil velocity of the rifle.

Show solution

Everything starts at rest, so the total momentum is zero before and after. Taking the bullet's direction as positive:

$$0 = 0.010 \times 400 + 4.0\,v \quad \Rightarrow \quad v = -\frac{4.0}{4.0} = -1.0 \text{ m s}^{-1}$$

The rifle moves back at 1.0 m s⁻¹. The minus sign tells you the direction, so keep it in your answer or state "backwards".

WORKED EXAMPLE

A railway truck of mass 12 000 kg moving at 3.0 m s⁻¹ hits a stationary truck of mass 8000 kg, and they couple together. The coupled trucks then roll 45 m before stopping. Calculate the average resistive force on them.

Show solution

Step 1, the collision (momentum):

$$12\,000 \times 3.0 = 20\,000\,v \quad \Rightarrow \quad v = 1.8 \text{ m s}^{-1}$$

Step 2, the deceleration (SUVAT): with $u = 1.8$, $v = 0$, $s = 45$:

$$0 = 1.8^2 + 2a(45) \quad \Rightarrow \quad a = -\frac{3.24}{90} = -0.036 \text{ m s}^{-2}$$

Step 3, the force (Newton's second law):

$$F = ma = 20\,000 \times 0.036 = 720 \text{ N}$$

Momentum is conserved during the brief collision, but afterwards the resistive force is an external force, so the momentum of the trucks falls to zero.

SUGGESTED EXPERIMENT

Investigating conservation of momentum with an air track

Two gliders sit on an air track, which floats them on a cushion of air so friction (an external force) is almost zero. Each glider carries a card of known length that passes through a light gate before and after the collision; velocity = card length ÷ time the gate is blocked. Weigh the gliders on a balance, then compare the total momentum before and after. Velcro or a pin and putty makes them stick; repelling magnets make them bounce. Level the track carefully, or a component of weight acts along it as an external force.

Quick checks

QUICK CHECK

A book rests on a table. Which force forms a Newton's third law pair with the weight of the book?

- The normal contact force of the table on the book
- The gravitational pull of the book on the Earth
- The normal contact force of the book on the table
- The weight of the table

The weight is the Earth pulling the book gravitationally, so its partner must be the book pulling the Earth gravitationally: different body, same type, equal and opposite. The table's push on the book acts on the same body as the weight, so it can't be the partner.

QUICK CHECK

A car travels along a straight, level road at a constant 25 m s^{-1}. Which statement is correct?

- The driving force is greater than the resistive forces.
- The resultant force on the car is zero.
- There are no forces acting on the car.
- The resultant force acts in the direction of motion.

Constant velocity means zero acceleration, so by Newton's first law the resultant force is zero. Plenty of forces act; they just cancel, with the driving force equal to the resistive forces.

QUICK CHECK

A lift is moving upwards and slowing down. Compared with the passenger's weight, the reading on bathroom scales in the lift is:

- greater
- less
- equal
- zero

Slowing down while going up means the acceleration is downwards, so a is negative in $R = m(g + a)$ and $R < mg$. The direction of the *velocity* doesn't matter; only the acceleration does.

QUICK CHECK

A skydiver falling at terminal velocity opens her parachute. Immediately afterwards:

- her velocity is upwards
- the resultant force on her is upwards, so she slows down
- the drag equals her weight
- her acceleration is 9.81 m s^{-2} downwards

The drag suddenly exceeds her weight, so the resultant force and the acceleration point up. She is still moving down, just more slowly each second, until the drag falls back to equal her weight.

QUICK CHECK

A stationary firework explodes into two pieces. Straight after the explosion, which statement is true?

- The total momentum is zero and the total kinetic energy is zero.
- The total momentum is zero and the total kinetic energy has increased.
- The total momentum and the total kinetic energy have both increased.
- Momentum is not conserved, because energy was released.

Momentum is conserved, so the total stays at zero: the pieces have equal and opposite momenta. Kinetic energy is a scalar and can't cancel; it comes from the chemical energy store of the explosive.

QUICK CHECK

A 2.0 kg trolley moving at 3.0 m s⁻¹ to the right hits a 1.0 kg trolley moving at 4.0 m s⁻¹ to the left. They stick together. What is their velocity afterwards?

- 0.67 m s⁻¹ to the right
- 3.3 m s⁻¹ to the right
- 0.67 m s⁻¹ to the left
- zero

With right as positive: $2.0 \times 3.0 + 1.0 \times (-4.0) = 2.0 \text{ kg m s}^{-1}$, and $v = 2.0/3.0 = 0.67 \text{ m s}^{-1}$. Forgetting the minus sign gives 3.3 m s⁻¹.

AS · Unit 1 · WPH11 Mechanics & Materials

Moments

A force can make something turn as well as move. How strongly it turns depends on how big the force is and how far its line of action is from the pivot, which is why door handles sit far from the hinges and a long spanner undoes a stiff nut. This page covers moments, centre of gravity and the principle of moments: everything you need for beams, planks, ladders and see-saws.

The moment of a force

Try opening a door by pushing right next to the hinge. It barely moves. Push at the handle with the same force and it swings easily. Push on the handle but directly *towards* the hinge, and nothing happens at all. The turning effect of a force, its **moment**, depends on the force and on how far its **line of action** passes from the pivot.

Moment of a force

FORMULA SHEET

$$\text{moment of force} = Fx$$

F: force / N

x: perpendicular distance from the line of action of the force to the axis of rotation / m

moment / N m

DEFINITION

Moment of a force

The moment of a force about a point is the force multiplied by the **perpendicular distance** from that point to the **line of action** of the force.

The **line of action** is the straight line along which the force acts, extended as far as you like in both directions. The distance x is measured at right angles to it. That one word, *perpendicular*, is where most of the marks are. Moments also have a sense of rotation: each one is either **clockwise** or **anticlockwise** about the point you choose.

Forces at an angle

When the force isn't at right angles to the spanner (or door, or beam), there are two equivalent ways to find the moment. Both give the same answer.

Two ways to get the moment of a force at an angle

1. A force F acts at a distance d along a spanner from the pivot, at an angle θ to the spanner.

2. **Method 1: perpendicular distance.** The line of action passes the pivot at a perpendicular distance $x = d\sin\theta$ (right-angled triangle with hypotenuse d):

$$\text{moment} = Fx = F\left(d\sin\theta\right)$$

3. **Method 2: resolve the force.** Split F into a component along the spanner, $F\cos\theta$, and one perpendicular to it, $F\sin\theta$. The component along the spanner points straight through the pivot, so its moment is zero. Only the perpendicular part turns:

$$\text{moment} = \left(F\sin\theta\right)d$$

4. The two are the same product, just grouped differently:

$$\boxed{\text{moment} = Fd\sin\theta}$$

It is biggest when $\theta = 90^\circ$ (push at right angles) and zero when $\theta = 0$ (push along the spanner, through the pivot).

COMMON MISTAKE

Don't multiply the force by the distance to the point where it acts unless the force is at right angles to that distance. A force whose line of action passes **through** the pivot has zero moment about it, however large it is.

WORKED EXAMPLE

A mechanic pulls on the end of a 0.25 m spanner with a force of 80 N at 60° to the spanner. (a) Calculate the moment about the nut. (b) What force at right angles to the spanner, at the same point, would give the same moment?

Show solution

$$\text{(a)}\quad x = 0.25\sin 60^\circ = 0.217\ \text{m}, \qquad \text{moment} = 80 \times 0.217 = 17.3\ \text{N m}$$

$$\text{(b)}\quad F = \frac{17.3}{0.25} = 69.3\ \text{N}$$

Pulling at right angles does the same job with about 11 N less effort.

EXAM TIP

The unit of moment is the newton metre, N m. It has the same base units as the joule (kg m^2 s^{-2}), but a moment is not an energy, so never write a moment in J. Always state the point you are taking moments about, and whether each moment is clockwise or anticlockwise.

The principle of moments

A see-saw balances when the turning effects on each side cancel. That idea, generalised to any object and any point, is the principle of moments.

DEFINITION

Principle of moments

For a body in equilibrium, the sum of the clockwise moments about any point equals the sum of the anticlockwise moments about the same point.

KEY IDEA

The two conditions for equilibrium

An extended body is in equilibrium only if **both** of these are true:

1. The **resultant force is zero**, so it has no linear acceleration (Newton's first law).
2. The **resultant moment about any point is zero**, so it has no rotational acceleration (the principle of moments).

If the resultant force is zero, the total moment comes out the same about every point, so if it balances about one point it balances about all of them. You can choose whichever point makes the algebra easiest.

WORKED EXAMPLE

A child of weight 300 N sits 1.5 m from the pivot of a uniform see-saw, which is pivoted at its centre. (a) Where must a 450 N child sit to balance it? (b) The see-saw weighs 200 N. What is the upward force from the pivot?

Show solution

(a) Take moments about the pivot. The see-saw's own weight acts at the pivot, so it has no moment there:

$$300 \times 1.5 = 450 \times x \quad \Rightarrow \quad x = 1.0 \text{ m}$$

(b) The resultant force is zero too, so the pivot pushes up with the total weight: $300 + 450 + 200 = 950$ N.

EXAM TIP

Take moments about the point where an unknown force acts that you **don't** want yet. That force passes through the point, so it has zero moment and drops out of the equation, leaving one unknown to solve for. Then use "upward forces = downward forces" for the rest.

Centre of gravity

DEFINITION

Centre of gravity

The centre of gravity of a body is the point through which the whole weight of the body can be taken to act.

Every tiny piece of an object has its own weight, and each of those little weights has its own moment about any point you pick. The centre of gravity is the one point where you can put the **whole** weight and get exactly the same total moment. That's why, for moments, you can replace a beam by a single arrow at its centre of gravity.

- For a **uniform** object (the same material and thickness throughout), the centre of gravity is at the geometric centre: the middle of a uniform rod, the centre of a uniform disc.
- For a symmetrical object, it lies on every line of symmetry.
- It doesn't have to be inside the object: the centre of gravity of a ring is in the empty middle, and that of a boomerang is in the air between its arms.

Where is the centre of gravity of a loaded rod?

1. A rod has a weight W_1 whose centre of gravity is at distance x_1 from one end, and a load W_2 is fixed at distance x_2. We want the single point $\bar{x}$ where the total weight $W_1 + W_2$ gives the same moment.

2. Take moments about the end. The total moment of the separate weights must equal the moment of the total weight acting at $\bar{x}$:

$$W_1 x_1 + W_2 x_2 = (W_1 + W_2)\,\bar{x}$$

3 Rearrange:

$$\boxed{\bar{x} = \frac{W_1 x_1 + W_2 x_2}{W_1 + W_2}}$$

It is a weighted average of the positions. A 1 m rod of weight 4 N with a 1 N load on one end has its centre of gravity at $(4 \times 0.5 + 1 \times 1.0)/5 = 0.60$ m: shifted towards the load, but not all the way.

Finding the centre of gravity by experiment

SUGGESTED EXPERIMENT

Centre of gravity of an irregular rod

The spec suggests this one. Two methods:

1. **Balance it.** Rest the rod on a knife edge (or a triangular prism) and slide it until it balances. The centre of gravity is directly above the knife edge, because only then does the weight have zero moment about it. Read the position off a metre rule.
2. **Use moments.** If the rod is too long or awkward to balance alone, hang a known weight from one end and find the new balance point. Measure the distances from the knife edge and use the principle of moments to calculate where the rod's own centre of gravity is (you need the rod's weight from a balance).

Using a sharp knife edge reduces the uncertainty in where the pivot is. Approach the balance point from both sides and take the mean, and read the scale at eye level to avoid parallax.

WORKED EXAMPLE

A non-uniform rod 1.00 m long has a weight of 4.0 N. A 2.0 N weight is hung from the 0 cm end, and the rod then balances on a knife edge at the 40 cm mark. Where is the rod's centre of gravity?

Show solution

Take moments about the knife edge. The 2.0 N weight is 0.40 m to the left, so it gives an anticlockwise moment. The rod's weight must give an equal clockwise moment, so its centre of gravity is a distance d to the right:

$$2.0 \times 0.40 = 4.0 \times d \quad \Rightarrow \quad d = 0.20 \text{ m}$$

The centre of gravity is at the $40 + 20 = 60$ cm mark.

Irregular flat shapes: the plumb-line method

To find the centre of gravity of a flat, irregular card (a **lamina**), make a few small holes near its edge. Hang it from a pin through one hole so it can swing freely, and hang a **plumb line** (a weight on a thread) from the same pin. When the card stops, draw along the thread. Repeat from a second hole. The centre of gravity is where the lines cross; a line from a third hole is a check.

Why does it work? While the centre of gravity is to one side of the pin, the weight has a moment about the pin that swings the card back. The card can only be at rest when that moment is zero, which means the line of action of the weight passes through the pin: the centre of gravity is **directly below** the pin, on the plumb line.

Free-body diagrams of extended bodies

For a particle, it didn't matter where you drew each force. For an extended body it matters a great deal, because the position of each force sets its moment. The rules:

- Draw the **weight** at the centre of gravity, vertically down.
- Draw each **support force** where the contact is. A support under a beam pushes up. A **smooth** wall or surface can only push at right angles to itself (no friction).
- A rough floor can supply **friction** along its surface as well as a normal reaction.
- Then apply both equilibrium conditions: forces balance **and** moments balance.

WORKED EXAMPLE

A uniform plank 4.0 m long and of weight 200 N rests on supports at each end, A and B. A person of weight 600 N stands 1.0 m from A. Calculate the forces from the two supports.

Show solution

Take moments about A, so the unknown force at A drops out. The plank's weight acts at its centre, 2.0 m from A:

$$R_B \times 4.0 = 200 \times 2.0 + 600 \times 1.0 = 1000 \quad \Rightarrow \quad R_B = 250 \text{ N}$$

Forces balance vertically:

$$R_A = 200 + 600 - 250 = 550 \text{ N}$$

Check with moments about B: $R_A \times 4.0 = 200 \times 2.0 + 600 \times 3.0 = 2200$, so $R_A = 550$ N. The support nearer the person takes more of the load.

WORKED EXAMPLE

The same plank now rests on support A at one end and support B 3.0 m from A, so 1.0 m overhangs. How far from A can the 600 N person walk before the plank tips?

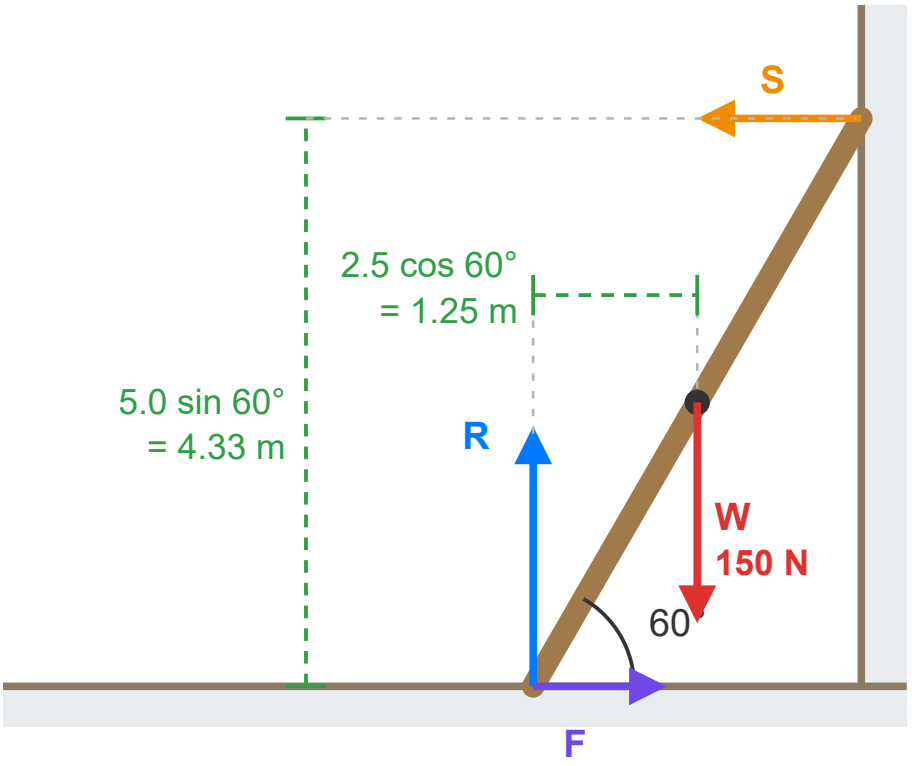
Show solution

Just as it is about to tip about B, the plank lifts off A, so $R_A = 0$. Take moments about B. The plank's weight acts 1.0 m on the A side of B; the person stands a distance $x - 3.0$ beyond B:

$$600\,(x - 3.0) = 200 \times 1.0 \quad \Rightarrow \quad x - 3.0 = 0.33\ \text{m}$$

The person can walk to 3.33 m from A, only a third of a metre past B. Try it in the simulation.

A ladder against a wall



A uniform 5.0 m ladder of weight 150 N leans on a smooth wall at 60° to the ground. The green dashed lines are the perpendicular distances from the foot of the ladder to the lines of action of S and W.

WORKED EXAMPLE

For the ladder in the diagram, calculate (a) the force S from the wall and (b) the normal reaction R and friction F at the ground.

Show solution

(a) Take moments about the **foot**, so that R and F (both acting there) drop out. S is horizontal, so its perpendicular distance from the foot is the height of the top of the ladder, $5.0\sin 60^\circ$. W is vertical, so its perpendicular distance is the horizontal distance to the midpoint, $2.5\cos 60^\circ$:

$$S \times 5.0\sin 60^\circ = 150 \times 2.5\cos 60^\circ$$

$$S \times 4.33 = 187.5 \quad \Rightarrow \quad S = 43.3\ \text{N}$$

(About the foot, W turns the ladder clockwise, towards the wall, and S turns it anticlockwise.)

(b) Resolve vertically: the only vertical forces are R and W, so $R = 150\ \text{N}$. Resolve horizontally: F must balance S, so $F = 43.3\ \text{N}$ towards the wall.

If the floor can't provide 43.3 N of friction, the foot slides out. Repeat the moments step for a general angle θ and you get $F = S = \dfrac{W}{2\tan\theta}$: the steeper the ladder, the less friction it needs, which is why ladders are safer at steep angles.

Quick checks

QUICK CHECK

The same 50 N force is applied to the end of a spanner in four different directions. Which gives the largest moment about the nut?

- Along the spanner, towards the nut
- At 30° to the spanner
- At 45° to the spanner
- At 90° to the spanner

The moment is $Fd\sin\theta$, largest when $\sin\theta = 1$. Pushing along the spanner sends the line of action through the nut, so the moment is zero.

QUICK CHECK

A uniform metre rule is pivoted at its 30 cm mark. It balances horizontally when a 0.20 kg mass hangs from the 10 cm mark. What is the mass of the rule?

- 0.067 kg
- 0.10 kg
- 0.20 kg
- 0.40 kg

The rule's weight acts at its centre, the 50 cm mark, 20 cm from the pivot. The mass is also 20 cm from the pivot on the other side, so the masses must be equal: $0.20g \times 0.20 = Mg \times 0.20$.

QUICK CHECK

Which statement must be true for an extended body in equilibrium?

- All the forces act through its centre of gravity.
- The resultant force is zero, but there can be a resultant moment.
- The resultant force is zero and the resultant moment about any point is zero.
- The clockwise moments are zero.

Equilibrium needs both conditions: no resultant force (no linear acceleration) and no resultant moment (no rotational acceleration). A couple is the classic case with zero resultant force but a non-zero moment, and it is not in equilibrium.

QUICK CHECK

A person walks along a plank that rests on two supports, from support A towards support B. What happens to the support forces?

- Both stay the same.
- The force at A decreases and the force at B increases; their sum stays the same.
- Both increase.
- The force at A increases and the force at B decreases.

Taking moments about A, R_B grows in proportion to the person's distance from A. The total weight doesn't change, so $R_A + R_B$ is constant and R_A must fall by the same amount.

QUICK CHECK

A card hung from a pin comes to rest with its centre of gravity directly below the pin. Why?

- Because the weight of the card acts at the pin.
- Only in that position does the weight have zero moment about the pin.
- Because the centre of gravity is always at the geometric centre.
- Because the pin pushes upwards harder in that position.

Anywhere else, the line of action of the weight misses the pin, so the weight has a moment about the pin that swings the card. It can only stay at rest when that moment is zero.

QUICK CHECK

A 20 N force acts on a door 0.80 m from the hinge. Its line of action is at 30° to the door. What is its moment about the hinge?

- 8.0 N m
- 13.9 N m
- 16.0 N m
- zero

The perpendicular distance is $0.80 \sin 30^\circ = 0.40 \text{ m}$, so the moment is $20 \times 0.40 = 8.0 \text{ N m}$. Using cos gives 13.9 N m, and ignoring the angle gives 16.0 N m.

AS · Unit 1 · WPH11 Mechanics & Materials

Energy

Energy never appears or disappears; it only moves from one store to another. Work done measures how much energy a force transfers, power measures how quickly it happens, and efficiency tells you how much ended up where you wanted it. With these few ideas you can solve problems that would be painful with forces and SUVAT alone.

Work done

In physics, **work** has a precise meaning: a force does work when its point of application moves in the direction of the force. Holding a heavy bag still is tiring, but your force on the bag does no work on it, because the bag doesn't move. Lifting it does.

Work done by a force

$$W = Fs$$

W: work done / J F: force / N s: distance moved in the direction of the force / m

DEFINITION

Work done and the joule

Work done = force × distance moved in the direction of the force. **1 joule** is the work done when a force of 1 N moves its point of application 1 m in the direction of the force, so $1\ \text{J} = 1\ \text{N m}$.

The big idea that makes work useful: **work done = energy transferred**. When you push a trolley and do 50 J of work on it, 50 J of energy leaves your chemical store and goes into the trolley's kinetic store (and some into thermal energy, if there's friction).

When the force is at an angle

Pull a sledge with a rope at an angle and only part of your pull moves it forwards. The rest just lifts the front of the sledge a little. Resolve the force along and perpendicular to the motion:

Work done by a force at an angle θ to the motion

1. The force F acts at angle θ to the direction of motion. Resolve it into a component along the motion and one at right angles:

$$F_{\parallel} = F\cos\theta \qquad\qquad F_{\perp} = F\sin\theta$$

2. The perpendicular component does no work: the sledge doesn't move in that direction at all. Only the component along the motion does work over the distance s:

$$\boxed{W = Fs\cos\theta}$$

At $\theta = 0$ this is just Fs. At $\theta = 90°$, $\cos\theta = 0$: a force at right angles to the motion does no work.

- **Zero work:** any force at right angles to the motion. The normal reaction on a car driving along a flat road, and the weight of a ball rolling across a level floor, do no work.
- **Negative work:** a force opposite to the motion ($\theta = 180°$, $\cos\theta = -1$), like friction or drag. It takes energy away from the object, usually into thermal energy.

WORKED EXAMPLE

A child pulls a sledge 25 m across flat snow with a force of 60 N along a rope at 30° to the horizontal. How much work does the pull do on the sledge?

Show solution

$$W = Fs\cos\theta = 60 \times 25 \times \cos 30° = 1300\ \text{J}\ (1.3\ \text{kJ})$$

If the sledge moves at a steady speed, its kinetic energy doesn't change, so all 1300 J is transferred to thermal energy by friction with the snow.

You do need the area idea for one case: the area under a force–extension graph is the elastic strain energy stored in a stretched material (see Stretching Materials).

Kinetic energy

A moving object has energy in its **kinetic store**. To find how much, ask how much work you would have to do to get it moving from rest.

Kinetic energy

$$E_k = \frac{1}{2}mv^2$$

E_k: kinetic energy / J m: mass / kg v: speed / m s^{-1}

Deriving $E_k = \frac{1}{2}mv^2$ from $W = Fs$

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 A constant resultant force F pushes a mass m, starting from rest, through a distance s. The work done on it is

$$W = Fs$$

2 Newton's second law gives $F = ma$, so

$$W = mas$$

3 The acceleration is constant, so use $v^2 = u^2 + 2as$ with $u = 0$:

$$v^2 = 2as \quad \Rightarrow \quad as = \frac{1}{2}v^2$$

4 Substitute for as. All the work done has gone into the kinetic store:

$$W = m\left(\frac{1}{2}v^2\right) \quad \Rightarrow \quad \boxed{E_k = \frac{1}{2}mv^2}$$

Why v^2? To reach a higher speed with the same force you must push for longer (time $\propto v$), and the object is also moving faster on average while you push (average speed $= \frac{1}{2}v$). The distance, and so the work, grows with both: $s = \frac{1}{2}v \times t \propto v^2$. That average speed is also where the ½ comes from.

The v^2 has real consequences. Double the speed and the kinetic energy goes up **four** times. To stop a car, the brakes must do work equal to its kinetic energy, $Fd = \frac{1}{2}mv^2$. With the same braking force, the braking distance is proportional to v^2: from 30 mph to 60 mph the braking distance doesn't double, it quadruples.

WORKED EXAMPLE

A car of mass 1200 kg travels at 20 m s⁻¹. The brakes can supply a constant braking force of 6000 N. (a) Calculate its kinetic energy. (b) Find its braking distance. (c) What would the braking distance be at 40 m s⁻¹?

Show solution

$$\text{(a)}\quad E_k = \tfrac{1}{2}(1200)(20)^2 = 2.4 \times 10^5 \text{ J}$$

$$\text{(b)}\quad Fd = E_k \quad \Rightarrow \quad d = \frac{2.4 \times 10^5}{6000} = 40 \text{ m}$$

(c) Double the speed gives four times the kinetic energy, so with the same force the distance is $4 \times 40 = 160$ m.

Gravitational potential energy

Lift something up and you do work against its weight. That energy isn't lost: it is stored in the **gravitational potential energy** store, ready to come back out as kinetic energy when the object falls.

Change in gravitational potential energy near the Earth's surface

$$\Delta E_{grav} = mg\Delta h$$

ΔE_{grav}: change in GPE / J m: mass / kg g = 9.81 N kg⁻¹ Δh: change in height / m

Deriving the change in GPE, ΔE = mgΔh, from W = Fs

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Lift a mass m at a steady speed, so its kinetic energy doesn't change. With no acceleration, your upward force exactly balances its weight:

$$F = mg$$

2. You move that force through a height Δh in its own direction, so the work done is

$$W = F\Delta h = mg\,\Delta h$$

3 All of that work is stored as gravitational potential energy:

$$\boxed{\Delta E_{grav} = mg\Delta h}$$

> *This assumes g is constant, which is true near the Earth's surface. Over large distances g falls off, and you need the field ideas in Unit 5.*

- Only **changes** in GPE matter. You can choose any level as zero (the floor, the table top, the ground) as long as you stick with it.
- Δh is the **vertical** height change. Walk up a long gentle ramp or climb a short steep ladder to the same height and the GPE gain is identical: the ramp needs a smaller force over a longer distance.

WORKED EXAMPLE

A 65 kg climber climbs from a ledge 12 m up a cliff to the top, which is 30 m above the ground. How much gravitational potential energy does she gain?

Show solution

Only the change in height counts: $\Delta h = 30 - 12 = 18 \text{ m}$.

$$\Delta E_{grav} = mg\Delta h = 65 \times 9.81 \times 18 = 1.15 \times 10^4 \text{ J}$$

Conservation of energy

DEFINITION

Principle of conservation of energy

Energy cannot be created or destroyed. It can only be transferred from one store to another, so the total energy of a closed system stays constant.

For mechanics problems this becomes a bookkeeping rule. As an object moves, energy shifts between its kinetic and gravitational stores, and any work done against friction or drag is transferred to thermal energy in the surroundings:

$$\text{loss in } E_{grav} = \text{gain in } E_k + \text{work done against resistive forces}$$

With no friction, all the GPE lost becomes KE: $mg\Delta h = \frac{1}{2}mv^2$, so $v = \sqrt{2g\Delta h}$. The mass cancels, which is why a heavy skater and a light skater released from the same height reach the bottom with the same speed. Notice that this method doesn't care about the **shape** of the track, only the change in height. That's what makes energy so powerful: you'd need calculus to do this with forces.

WORKED EXAMPLE

A roller-coaster car of mass 500 kg starts from rest at the top of a slope 32 m above the ground and descends to a point 12 m above the ground. (a) Ignoring friction, find its speed there. (b) Its actual speed there is 17.0 m s^{-1}. How much work was done against resistive forces? (c) The track between the two points is 120 m long. Find the average resistive force.

Show solution

(a) It falls $\Delta h = 20\ \text{m}$. All the GPE lost becomes KE:

$$\tfrac{1}{2}mv^2 = mg\Delta h \quad \Rightarrow \quad v = \sqrt{2 \times 9.81 \times 20} = 19.8\ \text{m s}^{-1}$$

(b) Now do the full energy bookkeeping:

$$\Delta E_{grav} = 500 \times 9.81 \times 20 = 98\,100\ \text{J}, \qquad E_k = \tfrac{1}{2} \times 500 \times 17.0^2 = 72\,250\ \text{J}$$

$$\text{work done against resistive forces} = 98\,100 - 72\,250 = 2.59 \times 10^4\ \text{J}$$

(c) Work done = force × distance, so

$$F = \frac{2.59 \times 10^4}{120} = 215\ \text{N}$$

Bouncing balls

A dropped ball turns GPE into KE on the way down. During the bounce it squashes, storing energy elastically, then springs back. But some energy is transferred to thermal energy in the ball and the floor, and to sound, so it rebounds with less KE and can't reach its starting height. The fraction of energy kept in each bounce is the same as the ratio of rebound height to drop height.

WORKED EXAMPLE

A tennis ball of mass 0.058 kg is dropped from 2.0 m and rebounds to 1.3 m. (a) Find its speed just before it hits the ground. (b) How much energy is transferred to other stores in the bounce? (c) What percentage of its energy does it keep?

Show solution

$$\text{(a)}\quad v = \sqrt{2g\Delta h} = \sqrt{2 \times 9.81 \times 2.0} = 6.3 \text{ m s}^{-1}$$

$$\text{(b)}\quad mg(2.0 - 1.3) = 0.058 \times 9.81 \times 0.70 = 0.40 \text{ J}$$

(c) Its GPE at the top of each bounce is proportional to the height, so it keeps $1.3/2.0 = 0.65$, or 65%, of its energy.

COMMON MISTAKE

Energy is never "used up" or "lost" in the sense of disappearing. When a question says "energy lost", it means energy transferred to stores you can't easily use, usually thermal energy of the surroundings. Your final answer should still balance: energy in = energy out.

Power

Two cranes can lift the same load to the same height, doing the same work. The more powerful one does it faster. **Power** is the rate of transferring energy, or the rate of doing work.

Power from energy transferred

$$P = \frac{E}{t}$$

P: power / W E: energy transferred / J
t: time / s

Power from work done

$$P = \frac{W}{t}$$

W: work done / J t: time / s

DEFINITION

Power and the watt

Power is the rate of energy transfer (or the rate of doing work). **1 watt** is a rate of energy transfer of 1 joule per second: $1 \text{ W} = 1 \text{ J s}^{-1}$.

COMMON MISTAKE

Don't confuse the symbols: W for work done (in J) and W for the watt (the unit of power). In $P = W/t$, the W is work. And watch for kW: 1 kW = 1000 W.

WORKED EXAMPLE

A crane lifts a load of mass 800 kg through a height of 15 m in 20 s at a steady speed. (a) Calculate the useful power output. (b) The motor's input power is 9.0 kW. Calculate its efficiency.

Show solution

(a) At steady speed, the useful work done is the gain in GPE:

$$P = \frac{mg\Delta h}{t} = \frac{800 \times 9.81 \times 15}{20} = 5890 \text{ W } (5.9 \text{ kW})$$

$$\text{(b)} \quad \text{efficiency} = \frac{\text{useful power output}}{\text{total power input}} = \frac{5890}{9000} = 0.65 = 65\%$$

Efficiency

No machine transfers all its input energy to where you want it. A motor heats up, its bearings rub, it hums. **Efficiency** measures the fraction of the input that is useful.

Efficiency (energy)

$$\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy input}}$$

Efficiency (power)

$$\text{efficiency} = \frac{\text{useful power output}}{\text{total power input}}$$

The two forms are the same thing: divide the top and bottom of the energy form by the same time t and you get the power form. Efficiency has no units. Give it as a decimal (0.65) or a percentage (65%), and it can never be more than 1, because that would mean creating energy. A **Sankey diagram** shows the same information as a picture: the width of each arrow is proportional to the energy or power it carries.

EXAM TIP

In efficiency questions, first decide what is **useful**. For a motor lifting a load it is the GPE gained, $mg\Delta h$; for a kettle it is the thermal energy of the water; for a light bulb it is the light. Then make sure the top and bottom of the fraction are both energies or both powers, never one of each.

Quick checks

QUICK CHECK

A waiter carries a tray across a room at a constant velocity, holding it level. How much work does the upward force of his hand do on the tray?

- It equals the tray's weight × the distance walked.
- Zero
- It equals the tray's kinetic energy.
- It depends on how fast he walks.

The force is vertical and the displacement is horizontal, so $\theta = 90^\circ$ and $W = Fs\cos 90^\circ = 0$. The tray's energy doesn't change: it stays at the same height and the same speed.

QUICK CHECK

A car's speed increases from 15 m s^{-1} to 30 m s^{-1}. With the same braking force, its braking distance becomes:

- the same
- twice as long
- four times as long
- eight times as long

Braking distance $d = \frac{1}{2}mv^2/F$. With m and F unchanged, $d \propto v^2$, so doubling v multiplies d by 4.

QUICK CHECK

A 1 kg ball and a 5 kg ball are released from rest at the same height on two frictionless slopes of different steepness. At the bottom, which is true?

- The 5 kg ball is faster.
- The ball on the steeper slope is faster.
- They have the same speed.
- The 1 kg ball is faster.

$mg\Delta h = \frac{1}{2}mv^2$ gives $v = \sqrt{2g\Delta h}$: the mass cancels and only the height change matters, not the shape of the slope. (The 5 kg ball does have five times the kinetic energy.)

QUICK CHECK

Which of these is equivalent to the watt in SI base units?

- kg m s^{-2}
- kg m^2 s^{-2}
- kg m^2 s^{-3}
- J s

1 W = 1 J s^{-1}, and 1 J = 1 N m = 1 kg m^2 s^{-2}. Dividing by seconds gives kg m^2 s^{-3}. The first option is the newton and the second is the joule.

QUICK CHECK

An electric motor with an input power of 40 W lifts a 2.0 kg mass through 3.0 m in 5.0 s. What is its efficiency?

- 15%
- 29%
- 68%
- 147%

Useful energy = $mg\Delta h = 2.0 \times 9.81 \times 3.0 = 58.9$ J. Energy input = $Pt = 40 \times 5.0 = 200$ J. Efficiency = 58.9 / 200 = 0.29. Comparing 58.9 J with 40 W gives the impossible 147%: never mix energy and power.

QUICK CHECK

A skater on a ramp with friction rises to a lower height on each swing. Where has the "missing" energy gone?

- It has been destroyed by friction.
- Mostly to thermal energy of the wheels, the ramp and the air.
- It has been converted into extra gravitational potential energy.
- It has been stored as kinetic energy in the ramp.

Work done against friction and drag transfers energy to thermal stores (and a little to sound). The total is conserved; it is just in a less useful form.

AS · Unit 1 · WPH11 Mechanics & Materials

Density, Upthrust & Viscous Drag

Why does a steel ship float when a steel bolt sinks? Why does a marble fall slowly through honey, and faster when the honey is warm? This topic is about how objects behave in fluids (liquids and gases). Three ideas do all the work: **density**, **upthrust** and **viscous drag**. Put them together and you can predict a ball's terminal velocity, which is exactly how Core Practical 2 measures the viscosity of a liquid.

Density

Density tells you how tightly mass is packed into a material: how many kilograms there are in each cubic metre.

Density

FORMULA SHEET

$$\rho = \frac{m}{V}$$

ρ: density / kg m⁻³ m: mass / kg V: volume / m³

A litre of water fills $1000\ \text{cm}^3 = 1.0 \times 10^{-3}\ \text{m}^3$ and has a mass of 1.0 kg, so its density is $1.0/1.0 \times 10^{-3} = 1000\ \text{kg m}^{-3}$. Density belongs to the **material**, not to the object: saw a steel bar in half and the mass and the volume both halve, so each half still has $\rho = 7800\ \text{kg m}^{-3}$.

DEFINITION

Density is mass per unit volume, $\rho = m/V$. Its SI unit is kg m⁻³.

Material	Density / kg m⁻³	Material	Density / kg m⁻³
air (20 °C)	1.2	sea water	1030
cork	≈ 240	glycerol	1260
pine wood	≈ 500	aluminium	2700
ice	917	steel	≈ 7800
vegetable oil	≈ 920	mercury	13 600
fresh water	1000	gold	19 300

COMMON MISTAKE

Volume units are the classic trap. $1\ \text{cm} = 10^{-2}\ \text{m}$, so $1\ \text{cm}^3 = (10^{-2})^3\ \text{m}^3 = 10^{-6}\ \text{m}^3$, and $1\ \text{mm}^3 = 10^{-9}\ \text{m}^3$. A handy check: $1\ \text{g cm}^{-3} = 1000\ \text{kg m}^{-3}$, so water is 1.00 g cm⁻³.

Measuring density

- **Regular solid:** mass on a balance; measure the sides with a ruler, vernier calipers or micrometer and calculate the volume.
- **Irregular solid:** mass on a balance; find the volume by **displacement**. Lower it into a measuring cylinder of water and read the rise in level, or use a displacement (Eureka) can and collect the water that overflows.
- **Liquid:** weigh an empty measuring cylinder, pour in a measured volume and weigh it again.

WORKED EXAMPLE

A crown has a mass of 1.20 kg. When it is lowered into a full displacement can, 75 cm³ of water overflows. The density of gold is 19 300 kg m⁻³. Is the crown pure gold?

Show solution

The crown displaces its own volume: $V = 75\ \text{cm}^3 = 75 \times 10^{-6}\ \text{m}^3$.

$$\rho = \frac{m}{V} = \frac{1.20}{75 \times 10^{-6}} = 1.6 \times 10^{4}\ \text{kg m}^{-3}$$

That is well below 19 300 kg m⁻³, so the crown is not pure gold; it has been mixed with a less dense metal. A pure gold crown of the same mass would displace only $1.20/19\,300 = 6.2 \times 10^{-5}\ \text{m}^3 = 62\ \text{cm}^3$. (This is the problem Archimedes is said to have solved in his bath.)

Upthrust

Push an empty plastic bottle under water and you can feel it shove back up. Every object in a fluid feels an upward force from that fluid. This force is called **upthrust**, and it obeys a beautifully simple rule.

Archimedes' principle

$$\text{upthrust} = \text{weight of fluid displaced} \qquad U = \rho_\text{f}\, V_\text{disp}\, g$$

U: upthrust / N ρ_f: density of the fluid / kg m^{-3}

V_disp: volume of fluid pushed aside (the submerged volume) / m^3 $g = 9.81\ \text{N kg}^{-1}$

The second form is just the first one written out: the displaced fluid has mass $\rho_\text{f} V_\text{disp}$ (from $\rho = m/V$), and its weight is that mass times g.

Why the rule works

Here is a lovely thought experiment. Take the object out and imagine the hole it leaves is filled with more of the fluid, the same shape as the object. That blob of fluid just sits there: it doesn't sink or rise, because it's simply part of the fluid. So the surrounding fluid must be pushing up on the blob with a force equal to the blob's **weight**.

Now put the object back. The surrounding fluid can't tell what is inside that surface: it pushes on the object in exactly the same way it pushed on the blob. So the upthrust on the object equals the weight of the fluid it has displaced, whatever the object is made of.

KEY IDEA

What upthrust does and doesn't depend on

- It depends on the **volume submerged** and the **density of the fluid**. Salt water gives more upthrust than fresh water.
- It does **not** depend on the object's mass or material. A steel ball and a plastic ball of the same size, both fully under water, feel the same upthrust.
- Once an object is fully submerged, pushing it deeper doesn't change the upthrust (liquids are almost incompressible, so the displaced volume stays the same).

DEFINITION

Upthrust is the upward force that a fluid exerts on an object that is wholly or partly immersed in it. It is equal to the weight of the fluid displaced.

Weighing things in water

Hang an object from a newton meter and lower it into water: the reading falls. The newton meter now only has to supply the part of the weight that the upthrust doesn't, so the reading is $W - U$, sometimes called the **apparent weight**. A displacement can lets you check Archimedes' principle directly: the water that overflows is the water displaced, so you can weigh it.

WORKED EXAMPLE

A stone hangs from a newton meter. The reading is 5.0 N in air and 3.2 N when the stone is completely submerged in water (density 1000 kg m^{-3}). Calculate (a) the upthrust, (b) the volume of the stone and (c) its density.

Show solution

(a) The newton meter reads $W - U$, so

$$U = 5.0 - 3.2 = 1.8 \text{ N}$$

(b) The upthrust is the weight of water displaced, and the stone displaces its own volume:

$$U = \rho_{\text{w}} V g \;\Rightarrow\; V = \frac{U}{\rho_{\text{w}} g} = \frac{1.8}{1000 \times 9.81} = 1.83 \times 10^{-4} \text{ m}^3$$

(c) Mass from the weight in air, then density:

$$m = \frac{W}{g} = \frac{5.0}{9.81} = 0.510 \text{ kg}, \qquad \rho = \frac{m}{V} = \frac{0.510}{1.83 \times 10^{-4}} = 2.8 \times 10^{3} \text{ kg m}^{-3}$$

Shortcut worth noticing: $W/U = \rho_{\text{stone}} V g / \rho_{\text{w}} V g = \rho_{\text{stone}}/\rho_{\text{w}}$, so $\rho_{\text{stone}} = 1000 \times 5.0/1.8 = 2800 \text{ kg m}^{-3}$ in one line.

EXAM TIP

Air is a fluid too, so everything in air feels a small upthrust. A party balloon 30 cm across displaces about 0.014 m³ of air, giving an upthrust of $1.2 \times 0.014 \times 9.81 \approx 0.17$ N. Fill it with helium (much less dense than air) and that is more than its weight, so it rises. For dense objects like a steel ball, upthrust in air is tiny and exam questions let you ignore it, but in a **liquid** you must include it.

Floating and sinking

Take an object of volume V and push it completely under. Its weight is $W = \rho_{\text{obj}} V g$ and the upthrust is $U = \rho_{\text{f}} V g$. The V and g are the same in both, so it all comes down to comparing densities:

- $\rho_{\text{obj}} > \rho_{\text{f}}$: $W > U$, so it **sinks**. On the bottom, the floor supplies the missing force: $W = U + R$, where R is the normal contact force.
- $\rho_{\text{obj}} < \rho_{\text{f}}$: $U > W$, so it **rises** until part of it sticks out. As it rises, the submerged volume shrinks and so does U, until $U = W$ and it **floats**.
- $\rho_{\text{obj}} = \rho_{\text{f}}$: it stays wherever you put it (neutral buoyancy, like a fish or a submarine).

How much of a floating object is under the surface?

1. A floating object is in equilibrium, so the resultant force on it is zero:

$$U = W$$

2. The upthrust is the weight of the fluid displaced by the submerged volume V_{sub}. The weight uses the whole volume V:

$$\rho_{\text{f}} V_{\text{sub}}\, g = \rho_{\text{obj}} V g$$

3. Cancel g and divide both sides by $\rho_{\text{f}} V$:

$$\boxed{\frac{V_{\text{sub}}}{V} = \frac{\rho_{\text{obj}}}{\rho_{\text{f}}}}$$

g cancels, so a boat floats at the same level on the Moon.

Ice (917 kg m⁻³) in sea water (1030 kg m⁻³) floats with $917/1030 = 0.89$ of its volume under the surface, which is why "the tip of the iceberg" is only about a tenth of it. A steel ship floats because what matters is its **average** density: the hull encloses a huge volume of air, so the ship's total mass divided by the volume of the hull is less than 1030 kg m⁻³. Load it with cargo and it sits lower in the water, displacing more water to balance the extra weight.

WORKED EXAMPLE

A rectangular block of wood measures 0.50 m × 0.20 m × 0.10 m. It floats in water (density 1000 kg m⁻³) with its largest face horizontal and 6.5 cm of its 10 cm thickness under the surface. (a) Calculate the density of the wood. (b) Calculate the smallest mass that could be placed on top of the block to push it completely under.

Show solution

(a) The cross-section is the same all the way down, so the fraction of the volume submerged is the fraction of the thickness submerged:

$$\frac{\rho_{\text{wood}}}{\rho_{\text{w}}} = \frac{6.5}{10} \;\Rightarrow\; \rho_{\text{wood}} = 0.65 \times 1000 = 650 \text{ kg m}^{-3}$$

(b) Volume $V = 0.50 \times 0.20 \times 0.10 = 0.010 \text{ m}^3$. Fully submerged, the upthrust and the block's weight are

$$U = 1000 \times 0.010 \times 9.81 = 98.1 \text{ N}, \qquad W = 650 \times 0.010 \times 9.81 = 63.8 \text{ N}$$

The extra weight needed is $98.1 - 63.8 = 34.3$ N, so the mass is

$$m = \frac{34.3}{9.81} = 3.5 \text{ kg}$$

Viscosity and Stokes' law

Viscosity is how "thick" a fluid is: its resistance to flowing. Honey has a high viscosity and water a low one. When a fluid flows, neighbouring layers slide past each other, and viscosity is like an internal friction between those layers. Its symbol is η (eta).

Now drop a ball into the fluid. The layer of fluid touching the ball is dragged along with it, the next layer out is dragged a little less, and so on: the ball has to keep shearing the fluid around it. That takes a force, so the fluid pulls back on the ball. This is **viscous drag**. For a small sphere moving slowly, George Stokes worked out exactly how big it is.

Stokes' law: viscous drag on a sphere

FORMULA SHEET

$$F = 6\pi\eta r v$$

F: viscous drag force / N η: viscosity of the fluid / Pa s r: radius of the sphere / m
v: speed of the sphere relative to the fluid / m s⁻¹

Why each factor is there

- η: a stickier fluid resists the shearing more.
- v: moving faster means the layers shear past each other faster. At low speeds the drag is simply proportional to the speed.
- r: a bigger sphere drags a bigger region of fluid along with it.
- 6π: this number comes from the geometry of a sphere. Working it out needs maths well beyond A level, so just use it.

Rearranging gives the unit of viscosity: $\eta = F/(6\pi r v)$ has units $\text{N}/(\text{m} \times \text{m s}^{-1}) = \text{N s m}^{-2}$. Since $1 \text{ N m}^{-2} = 1 \text{ Pa}$, that is the **pascal second, Pa s**, which in base units is kg m⁻¹ s⁻¹.

KEY IDEA

When can you use Stokes' law?

Only for **small**, **spherical** objects moving at **low speeds** with **laminar flow** (no turbulence) around them. A raindrop in air or a ball bearing in thick oil is fine; a football, a car or a skydiver is not. And because viscosity is **temperature dependent**, the value of η you use must be the one for the temperature of the fluid.

Viscosity depends on temperature

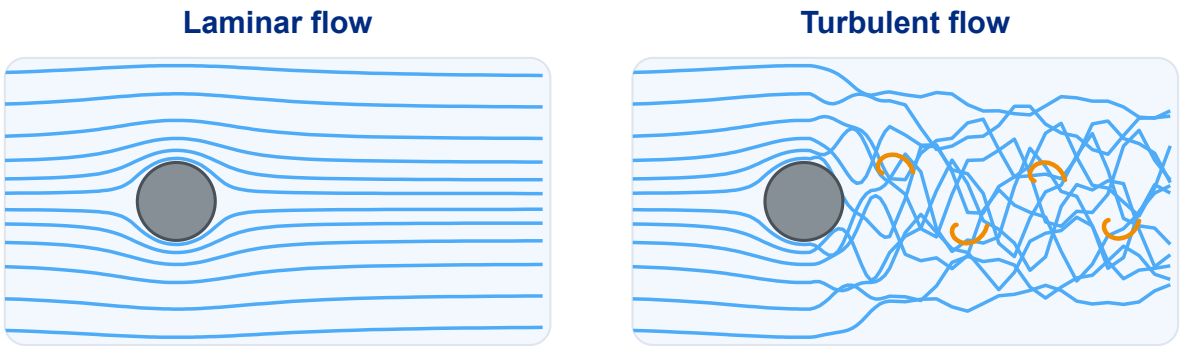
For liquids, viscosity falls steeply as the temperature rises. Golden syrup pours easily after the tin has stood in hot water, and cold engine oil is so thick that engines wear fastest just after starting. Glycerol is about **five times** less viscous at 40 °C than at 20 °C. So any viscosity value only means something if you also give the temperature, and in an experiment you must measure the temperature and keep it steady.

Fluid (at 20 °C)	Viscosity η / Pa s
air	1.8×10^{-5}
water	1.0×10^{-3}
olive oil	≈ 0.08
castor oil	≈ 1.0
glycerol	≈ 1.4
honey	roughly 2 to 10 (depends on the honey)

Laminar and turbulent flow

To picture a flow we draw **streamlines**: lines that show the path a small piece of the fluid follows. Where the streamlines squeeze closer together, the fluid is moving faster (the same amount of fluid is getting through a narrower gap).

- In **laminar flow** the fluid moves in smooth layers that slide past each other without mixing. The streamlines are smooth and never cross, and the velocity at any point stays the same over time.
- In **turbulent flow** the fluid swirls into eddies and mixes. The flow pattern keeps changing, and the speed and direction at any point change erratically.



Fluid flowing from left to right past a sphere. Laminar flow (left): smooth, non-crossing streamlines, bunched up where the fluid speeds past the sides of the sphere. Turbulent flow (right): behind the sphere the flow breaks into eddies and the layers mix.

DEFINITION

Laminar flow: the fluid flows in layers with no mixing between them; streamlines do not cross and the velocity at each point is constant.

Turbulent flow: the fluid mixes in irregular eddies; the speed and direction of flow at each point change with time.

Flow round an object tends to be laminar when the object is **small**, moving **slowly** and the fluid is **viscous**. Speed it up and a wake of eddies forms behind it. Stokes' law assumes the smooth, layered flow, so it only works in the laminar case. Once there is turbulence the object also has to keep setting the fluid swirling, which costs extra energy, so the drag becomes bigger than $6\pi\eta r v$.

EXAM TIP

A favourite question: "Explain why Stokes' law cannot be used to find the drag on a skydiver." Give the conditions it breaks: a skydiver is not small, not spherical and moves fast, so the air flow around them is turbulent.

Terminal velocity

Let's put the three ideas together. A small sphere of radius r and density ρ_{s} is released from rest in a liquid of density ρ_{f} and viscosity η. Three forces act on it:

- **Weight** $W = mg = \frac{4}{3}\pi r^3 \rho_{\mathrm{s}} g$, downwards.
- **Upthrust** $U = \frac{4}{3}\pi r^3 \rho_{\mathrm{f}} g$, upwards (the weight of liquid displaced).
- **Viscous drag** $F = 6\pi\eta r v$, upwards (always opposite to the velocity).

1. At the instant of release $v = 0$, so there is no drag. The resultant force is $W - U$ and the ball accelerates downwards (with less than g, because of the upthrust).
2. As the ball speeds up, the drag grows in proportion to v. The resultant force $W - U - F$ shrinks, so the acceleration gets smaller and smaller.
3. Eventually F has grown so that $W = U + F$. The resultant force is zero, so (by Newton's first law) the ball carries on at a constant velocity: its **terminal velocity**.

On a velocity–time graph this gives a curve that starts steep (gradient = initial acceleration) and flattens off towards a horizontal line at the terminal velocity.

DEFINITION

Terminal velocity is the constant, maximum velocity of an object falling through a fluid, reached when the resultant force on it is zero (for a sphere in a liquid, weight = upthrust + viscous drag).

Terminal velocity of a sphere falling through a liquid

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 At terminal velocity the resultant force is zero, so the upward forces balance the weight:

$$W = U + F$$

2 Write the weight using the density of the sphere. Its volume is $\frac{4}{3}\pi r^3$, so its mass is $\frac{4}{3}\pi r^3 \rho_\mathrm{s}$:

$$W = \tfrac{4}{3}\pi r^3 \rho_\mathrm{s}\, g$$

3 The upthrust is the weight of the liquid displaced, which has the same volume but the liquid's density:

$$U = \tfrac{4}{3}\pi r^3 \rho_\mathrm{f}\, g$$

4 The drag at terminal velocity v_t comes from Stokes' law:

$$F = 6\pi\eta r v_\mathrm{t}$$

5 Substitute all three into $W = U + F$ and collect the two "volume" terms:

$$6\pi\eta r v_\mathrm{t} = \tfrac{4}{3}\pi r^3 g\,(\rho_\mathrm{s} - \rho_\mathrm{f})$$

$\rho_\mathrm{s} - \rho_\mathrm{f}$ appears because the upthrust "cancels out" part of the weight.

6 Divide both sides by $6\pi\eta r$. The π cancels and one power of r cancels, and $\frac{4}{3} \div 6 = \frac{4}{18} = \frac{2}{9}$:

$$\boxed{v_\mathrm{t} = \frac{2r^2 g\,(\rho_\mathrm{s} - \rho_\mathrm{f})}{9\eta}}$$

7 Rearranged for the viscosity, which is what Core Practical 2 uses:

$$\eta = \frac{2r^2 g\,(\rho_\mathrm{s} - \rho_\mathrm{f})}{9v_\mathrm{t}}$$

The result tells you a lot at a glance:

- $v_\mathrm{t} \propto r^2$: double the radius and the terminal velocity goes up **four** times. The weight (minus upthrust) grows with r^3 but the drag only with r, so big balls win.
- $v_\mathrm{t} \propto 1/\eta$: warm the liquid, its viscosity falls, and balls fall faster.
- If $\rho_\mathrm{s} < \rho_\mathrm{f}$ the formula gives a negative answer: the sphere rises at a steady speed instead, like a bubble in a fizzy drink.

WORKED EXAMPLE

A steel ball bearing of radius 1.0 mm falls at a terminal velocity of 1.2 cm s⁻¹ through an oil of density 960 kg m⁻³. The density of steel is 7800 kg m⁻³. (a) Calculate the weight of the ball and the upthrust on it. (b) Calculate the viscosity of the oil. (c) State one assumption you have made.

Show solution

(a) Volume $V = \frac{4}{3}\pi(1.0 \times 10^{-3})^3 = 4.19 \times 10^{-9}\ \mathrm{m}^3$.

$$W = \rho_\mathrm{s} V g = 7800 \times 4.19 \times 10^{-9} \times 9.81 = 3.21 \times 10^{-4}\ \mathrm{N}$$

$$U = \rho_\mathrm{f} V g = 960 \times 4.19 \times 10^{-9} \times 9.81 = 3.95 \times 10^{-5}\ \mathrm{N}$$

(b) At terminal velocity $F = W - U = 3.21 \times 10^{-4} - 0.39 \times 10^{-4} = 2.81 \times 10^{-4}\ \mathrm{N}$. From Stokes' law:

$$\eta = \frac{F}{6\pi r v} = \frac{2.81 \times 10^{-4}}{6\pi \times 1.0 \times 10^{-3} \times 0.012} = 1.2\ \mathrm{Pa\ s}$$

(The formula $\eta = 2r^2 g(\rho_\mathrm{s} - \rho_\mathrm{f})/9v_\mathrm{t}$ gives the same answer in one step.)

(c) Any one of: the flow round the ball is laminar (so Stokes' law applies); the ball is a perfect sphere; the ball had reached terminal velocity; the temperature of the oil was constant; the walls of the container don't affect the motion.

COMMON MISTAKE

Don't write "at terminal velocity, weight = drag". For an object in a liquid the upthrust is a significant part of the balance: weight = upthrust + drag. Only drop the upthrust when the fluid is a gas and the object is much denser than it.

Core Practical 2: viscosity by the falling-ball method

CORE PRACTICAL

Use a falling-ball method to determine the viscosity of a liquid

A tall measuring cylinder is filled with a viscous, transparent liquid such as glycerol or washing-up liquid. Rubber bands round the cylinder mark out timing points. Small steel ball bearings are dropped in and timed between the bands.

1. Measure the **diameter** of each ball with a **micrometer** at several orientations and average; halve it to get r.
2. Find the densities: ρ_s from the mass and volume of a number of balls, and ρ_f by weighing a measured volume of the liquid (or use given values).
3. Place the top band well below the surface, so the ball is at terminal velocity before timing starts. Measure the distance d between bands with a metre rule, with your eye level with each band.
4. Drop the ball gently in the centre of the cylinder (tweezers, or a short tube held just above the surface). Time it between the bands with a stopwatch, or film it beside a ruler and step through the video frame by frame. Then $v = d/t$.
5. Check terminal velocity has been reached: use three equally spaced bands and confirm the two time intervals are the same.
6. Repeat for each ball size, measure the temperature of the liquid with a thermometer, and keep it constant.

Graph: $v_\mathrm{t} = \dfrac{2g(\rho_\mathrm{s} - \rho_\mathrm{f})}{9\eta}\, r^2$ has the form $y = mx$. Plot v against r^2: a straight line through the origin with

$$\text{gradient} = \frac{2g(\rho_\mathrm{s} - \rho_\mathrm{f})}{9\eta} \quad \Rightarrow \quad \eta = \frac{2g(\rho_\mathrm{s} - \rho_\mathrm{f})}{9 \times \text{gradient}}$$

Sources of uncertainty and how to reduce them

- **Reaction time** when starting and stopping the stopwatch: use bands far apart so the times are long, repeat and average, or use video analysis or light gates.

- **Parallax** when judging when the ball passes a band: keep your eye level with the band.
- **Radius**: r is squared, so its percentage uncertainty is doubled. Use a micrometer, not a ruler.
- **Temperature**: η changes a lot with temperature, so measure it and avoid warming the liquid (don't hold the cylinder).
- **Walls**: a ball close to the wall of a narrow tube is slowed down. Use a wide cylinder and drop balls down the middle.
- **Air bubbles** sticking to the ball add upthrust: wet the balls in the liquid before dropping them.

Safety: wipe up spills straight away (glycerol makes the floor slippery), clamp the tall cylinder so it can't topple, and retrieve the balls with a magnet.

WORKED EXAMPLE

In a falling-ball experiment a student plots terminal velocity v against r^2 for steel balls (density 7800 kg m⁻³) falling through a liquid of density 1260 kg m⁻³. The best-fit line passes through the origin and has a gradient of $7.6 \times 10^3\ \text{m}^{-1}\ \text{s}^{-1}$. (a) Calculate the viscosity of the liquid. (b) The student's value is higher than the value in a data book. Suggest one reason why.

Show solution

(a) The gradient is $2g(\rho_\text{s} - \rho_\text{f})/9\eta$, so

$$\eta = \frac{2g(\rho_\text{s} - \rho_\text{f})}{9 \times \text{gradient}} = \frac{2 \times 9.81 \times (7800 - 1260)}{9 \times 7.6 \times 10^3} = 1.9\ \text{Pa s}$$

Check the units: $\text{m s}^{-2} \times \text{kg m}^{-3}/(\text{m}^{-1}\,\text{s}^{-1}) = \text{kg m}^{-1}\,\text{s}^{-1} = \text{Pa s}$.

(b) The liquid was colder than the data-book temperature (viscosity rises as temperature falls); or the walls of a narrow cylinder slowed the balls, making v too small and η too big.

Quick checks

QUICK CHECK

A block floats in water (density 1000 kg m⁻³) with three-quarters of its volume below the surface. What is the density of the block?

- 250 kg m⁻³
- 750 kg m⁻³
- 1000 kg m⁻³
- 1333 kg m⁻³

For a floating object $V_{\text{sub}}/V = \rho_{\text{obj}}/\rho_{\text{f}}$, so $\rho_{\text{obj}} = 0.75 \times 1000 = 750\ \text{kg m}^{-3}$. 1333 comes from dividing the wrong way round; a floating object can't be denser than the liquid.

QUICK CHECK

A steel ball is fully submerged in water and then lowered deeper. What happens to the upthrust on it?

- It increases, because the pressure is greater deeper down.
- It decreases, because there is more water above the ball pushing down.
- It stays the same, because the volume of water displaced is unchanged.
- It is zero, because steel is denser than water.

Upthrust is the weight of fluid displaced, and a fully submerged ball always displaces its own volume. The pressure is bigger on both its top and its bottom, but the *difference* between them stays the same. A sinking object still feels upthrust; it just isn't enough to balance its weight.

QUICK CHECK

Which of these is a correct unit for viscosity?

- N m s^{-1}
- kg m s^{-1}
- kg m^{-1} s^{-1}
- N m^{-1}

$\eta = F/6\pi r v$ has units N/(m × m s^{-1}) = N s m^{-2} = Pa s. Writing N as kg m s^{-2} gives kg m s^{-2} × s × m^{-2} = kg m^{-1} s^{-1}.

QUICK CHECK

Two steel balls fall through the same glycerol. Ball Q has twice the radius of ball P. Stokes' law applies to both. What is the ratio (terminal velocity of Q) / (terminal velocity of P)?

- 2
- 4
- 8
- 1, because all objects fall at the same rate

$v_t \propto r^2$. Doubling r makes the weight minus upthrust 8 times bigger (it goes as r^3), but the drag at a given speed only doubles (it goes as r), so the speed must be 4 times bigger for the forces to balance. "Everything falls at the same rate" is only true with no drag and no upthrust.

QUICK CHECK

The glycerol in a falling-ball experiment warms up from 18 °C to 25 °C during the lesson. What happens to the terminal velocity of a ball bearing?

- It decreases, because warm liquids are more viscous.
- It increases, because the viscosity of the glycerol decreases.
- It stays the same, because the densities have not changed.
- It increases, because the upthrust on the ball decreases a lot.

The viscosity of a liquid falls as its temperature rises, and $v_t \propto 1/\eta$. The density of glycerol does fall very slightly when it warms, but that effect is tiny compared with the large drop in viscosity.

QUICK CHECK

A small sphere is falling through oil at its terminal velocity. Which statement is correct?

- The viscous drag equals the weight of the sphere.
- The resultant force is downwards but too small to notice.
- The viscous drag equals the weight minus the upthrust.
- The upthrust equals the weight of the sphere.

At terminal velocity the resultant force is exactly zero: $W = U + F$, so $F = W - U$. If the upthrust equalled the weight, the sphere would not be falling at all.

AS · Unit 1 · WPH11 Mechanics & Materials

Stretching Materials

Hang a mass on a spring and it stretches; double the mass and it stretches twice as far. Pull a copper wire hard enough and it stays longer when you let go; pull a glass rod and it simply snaps. This topic gives you the language and the graphs to describe all of that: **Hooke's law** and **stiffness** for objects, **stress**, **strain** and the **Young modulus** for materials, and the **elastic strain energy** stored when something is stretched.

Hooke's law and stiffness

Hang a 100 g mass on a spring and it stretches by, say, 4 cm. Add another 100 g and it stretches another 4 cm. Every equal step in force gives an equal step in **extension**, so the extension is directly proportional to the force. That is Hooke's law.

Hooke's law

FORMULA SHEET

$$F = k\,\Delta x$$

F: force applied / N k: stiffness / N m⁻¹ Δx: extension (or compression) / m

The constant k is the **stiffness** of the object: the force needed per metre of extension. A stiff spring has a large k and barely stretches; a floppy one has a small k. On a graph of force (y-axis) against extension (x-axis), Hooke's law gives a straight line through the origin, and its **gradient is** k.

DEFINITION

Hooke's law: the extension of an object is directly proportional to the force applied to it, provided the limit of proportionality is not exceeded.

Stiffness k: the force per unit extension, $k = F/\Delta x$, measured in N m⁻¹.

Hooke's law works for squashing too. Push on the ends of a spring and its **compression** Δx (the decrease in length) is proportional to the force, with the same equation. Stiffness belongs to the **object**: a thicker wire, a shorter spring or a spring made of stiffer metal all have a bigger k. We'll fix that with the Young modulus in a moment.

COMMON MISTAKE

Extension is the **change** in length, not the new length. A spring with a natural length of 12.0 cm that is stretched to 15.0 cm has $\Delta x = 3.0\ \text{cm} = 0.030\ \text{m}$. And always convert cm or mm to metres before using $F = k\Delta x$.

WORKED EXAMPLE

A spring has a natural length of 80 mm. When a 300 g mass hangs from it, its length is 104 mm. (a) Calculate the stiffness of the spring. (b) Calculate the length of the spring when the load is increased to 450 g, assuming the limit of proportionality is not exceeded.

Show solution

(a) The force is the weight of the mass, and the extension is the change in length:

$$F = mg = 0.300 \times 9.81 = 2.94\ \text{N}, \qquad \Delta x = 104 - 80 = 24\ \text{mm} = 0.024\ \text{m}$$

$$k = \frac{F}{\Delta x} = \frac{2.94}{0.024} = 123\ \text{N m}^{-1}$$

(b) Extension is proportional to the load, so 450 g gives $\frac{450}{300} \times 24 = 36\ \text{mm}$ of extension. The length is $80 + 36 = 116\ \text{mm}$.

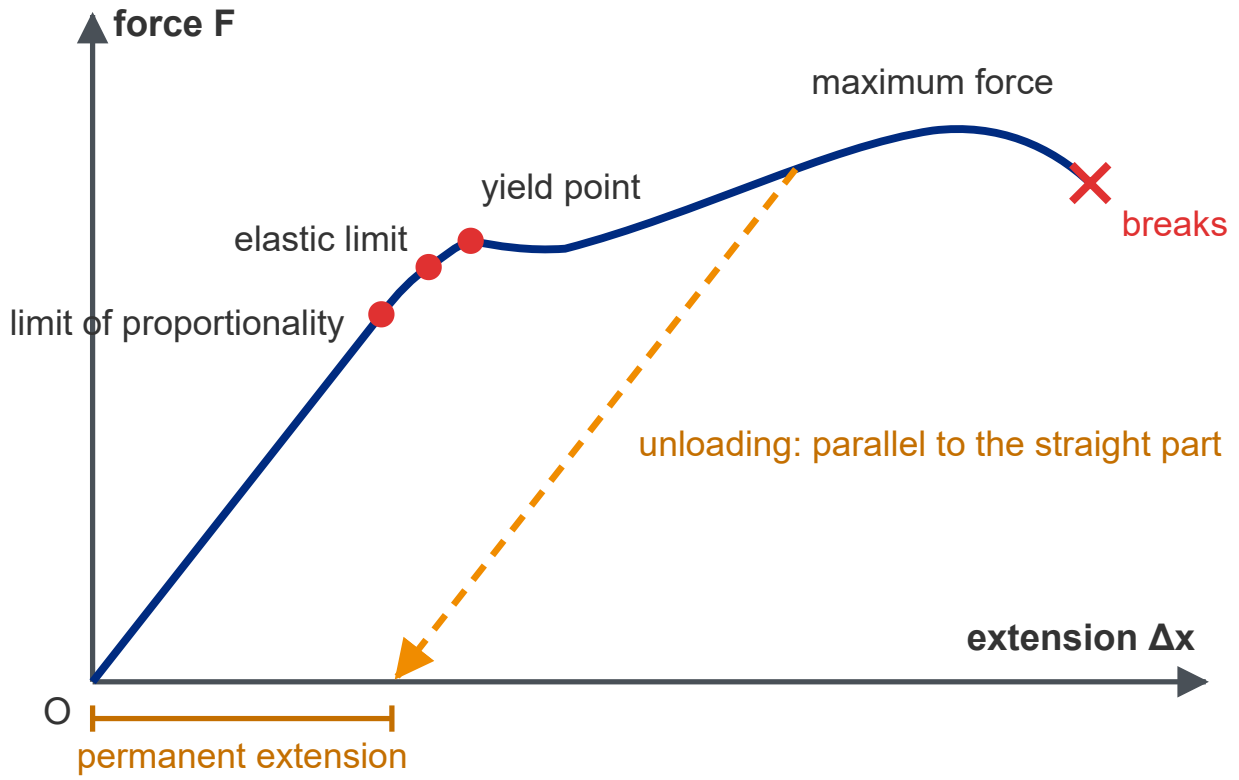
Using ratios like this avoids rounding errors, and it doesn't even need g.

EXAM TIP

Check the axes before you read a gradient. Edexcel usually puts force on the y-axis, so the gradient is k. If a graph has extension on the y-axis and force on the x-axis, the gradient is $1/k$, and a steeper line means a *less* stiff spring.

Force–extension graphs: elastic and plastic

Load a metal wire until it breaks and the force–extension graph tells the whole story. It starts as a straight line (Hooke's law), then bends over, and eventually the wire starts to stretch a lot for very little extra force before it finally breaks.



A metal wire stretched to breaking (not to scale). Up to the elastic limit the deformation is elastic. Beyond it the deformation is plastic: unloading follows a line parallel to the straight part and leaves a permanent extension.

DEFINITION

Limit of proportionality: the point beyond which force is no longer proportional to extension (Hooke's law stops applying).

Elastic limit: the maximum force (or extension) for which the object will still return to its original length when the force is removed.

Yield point: the point beyond which the material stretches a lot (plastically) with little or no increase in force.

Elastic deformation: the object returns to its original shape and size when the deforming force is removed.

Plastic deformation: the object does not return to its original shape when the force is removed; it is permanently deformed.

COMMON MISTAKE

The limit of proportionality and the elastic limit are **not** the same thing. Between them the graph is already curved, but the wire still springs back to its original length. Rubber is the extreme case: it is elastic up to huge extensions but its force–extension graph is curved almost everywhere.

Why the unloading line is parallel

Force–compression graphs

For a compressive force you plot force against **compression** and the same ideas apply: a straight section where Hooke's law holds (gradient = stiffness), and elastic or plastic behaviour beyond it. Squash a spring and the graph is straight until the coils touch; after that it shoots up almost vertically, because you are now compressing solid metal. Squash a lump of modelling clay and it deforms plastically almost straight away.

Stress, strain and the Young modulus

Stiffness describes one particular spring or wire. Engineers want a number that describes the **material**, whatever its shape. Two thought experiments show how to get one:

- Hang a load from **two identical wires side by side**. Each carries half the load, so each stretches half as much. Doubling the cross-sectional area halves the extension, so what matters is the **force per unit area**.
- Use a wire **twice as long**. Each half carries the full load and stretches just as much as a single short wire, so the total extension doubles. What matters is the **extension per unit length**.

Stress

FORMULA SHEET

$$\sigma = \frac{F}{A}$$

F: force / N A: cross-sectional area / m²

σ: stress / Pa

Strain

FORMULA SHEET

$$\varepsilon = \frac{\Delta x}{x}$$

Δx: extension / m x: original length / m

ε: strain (no unit)

Young modulus

FORMULA SHEET

$$E = \frac{\sigma}{\varepsilon}$$

E: Young modulus / Pa

valid in the Hooke's law region

Stress has the same unit as pressure, the pascal (1 Pa = 1 N m^{-2}); for metals you'll usually see MPa (10^6 Pa). **Strain** is a length divided by a length, so it has **no unit** (sometimes it's given as a percentage). That means the

Young modulus also has the unit Pa, usually quoted in GPa (10^9 Pa). The same definitions work for **tensile** (stretching) and **compressive** (squashing) forces.

DEFINITION

Stress: force per unit cross-sectional area, $\sigma = F/A$.

Strain: extension (or compression) per unit original length, $\varepsilon = \Delta x/x$.

Young modulus: the ratio of stress to strain, $E = \sigma/\varepsilon$, for a material obeying Hooke's law. It measures the stiffness of the material.

Linking stiffness to the Young modulus

1 Substitute the definitions of stress and strain into $E = \sigma/\varepsilon$:

$$E = \frac{F/A}{\Delta x/x} = \frac{F\,x}{A\,\Delta x}$$

This form is the one you use with raw measurements, for example in Core Practical 3.

2 Rearrange to get $F/\Delta x$, which is the stiffness:

$$\boxed{k = \frac{F}{\Delta x} = \frac{EA}{x}}$$

A thicker wire (bigger A) is stiffer, a longer wire (bigger x) is less stiff, and a stiffer material (bigger E) gives a stiffer wire, exactly as the thought experiments said.

Material	Young modulus / GPa	Material	Young modulus / GPa
rubber	0.01 to 0.1	aluminium	70
nylon	≈ 3	glass	≈ 70
bone (along its length)	≈ 17	titanium alloy	≈ 110
concrete	≈ 30	copper	≈ 120
wood (along the grain)	≈ 10	steel	≈ 200

These numbers matter in real design. A replacement hip joint ("spare-part surgery") has to be strong and must not be too much stiffer than the bone it is fixed into, which is one reason titanium alloys are used rather than steel. Food scientists test the "crunch" of a biscuit by compressing it and measuring the force, and the stiffness and breaking point tell them how it will feel to bite.

WORKED EXAMPLE

A steel wire of length 2.50 m and diameter 0.50 mm supports a load of 60 N. Its extension is 3.8 mm. Calculate (a) the stress, (b) the strain and (c) the Young modulus of the steel.

Show solution

(a) Cross-sectional area from the diameter ($r = 0.25$ mm):

$$A = \pi r^2 = \pi(0.25 \times 10^{-3})^2 = 1.96 \times 10^{-7}\ \text{m}^2$$

$$\sigma = \frac{F}{A} = \frac{60}{1.96 \times 10^{-7}} = 3.06 \times 10^{8}\ \text{Pa}$$

(b) $\varepsilon = \dfrac{\Delta x}{x} = \dfrac{3.8 \times 10^{-3}}{2.50} = 1.52 \times 10^{-3}$

(c) $E = \dfrac{\sigma}{\varepsilon} = \dfrac{3.06 \times 10^{8}}{1.52 \times 10^{-3}} = 2.0 \times 10^{11}\ \text{Pa}$ (200 GPa).

The classic slip is to use the diameter instead of the radius in πr^2, which makes the area four times too big.

WORKED EXAMPLE

When a 70 kg runner lands, the force on one thigh bone (femur) can be about three times their weight. The femur is 0.45 m long, with a cross-sectional area of 3.3 × 10⁻⁴ m² and a Young modulus of 17 GPa. Calculate the compressive stress and the amount the bone shortens.

Show solution

$$F = 3 \times 70 \times 9.81 = 2060\ \text{N}, \qquad \sigma = \frac{F}{A} = \frac{2060}{3.3 \times 10^{-4}} = 6.2 \times 10^{6}\ \text{Pa}$$

$$\varepsilon = \frac{\sigma}{E} = \frac{6.2 \times 10^{6}}{17 \times 10^{9}} = 3.7 \times 10^{-4}$$

$$\Delta x = \varepsilon x = 3.7 \times 10^{-4} \times 0.45 = 1.7 \times 10^{-4}\ \text{m} \approx 0.17\ \text{mm}$$

Tiny, and the stress is far below the breaking stress of bone (well over 100 MPa in compression), which is why bones survive running.

Stress–strain graphs and breaking stress

A stress–strain graph is a force–extension graph with the axes rescaled: divide the force by A and the extension by x. The shape is the same, but now the graph describes the **material**, not one particular sample. In the region where Hooke's law holds, the **gradient is the Young modulus**.

DEFINITION

Breaking stress: the stress at which a material breaks (fractures).

Different materials give very different curves. These words aren't spec definitions, but they are the natural way to describe what the graphs show:

- **Stiff:** a steep straight section (large E). Steel is stiffer than copper; rubber is not stiff at all.
- **Strong:** a high breaking stress.
- **Ductile:** a long plastic region, so it can be drawn out into wires. Copper is very ductile.
- **Brittle:** breaks with little or no plastic deformation. Glass obeys Hooke's law right up to the moment it snaps.
- **Rubber:** stretches to several times its length and still returns, but its graph is curved, so it never really obeys Hooke's law.

Mild steel shows a clear **yield point**: the stress suddenly stops rising (it even dips slightly) while the steel stretches a lot. After that the metal gets harder to stretch again (it "work hardens") until the maximum stress, and then it breaks. The spec only uses the term breaking stress; many books also call the maximum stress on the curve the ultimate tensile stress.

WORKED EXAMPLE

The stress–strain graph for a metal is a straight line from the origin to a stress of 240 MPa at a strain of 0.0020, after which it curves. (a) Calculate the Young modulus. (b) A wire of this metal is 2.0 m long with a cross-sectional area of 0.50 mm². Calculate the largest force for which it obeys Hooke's law, and the extension at that force.

Show solution

(a) $E = \dfrac{240 \times 10^{6}}{0.0020} = 1.2 \times 10^{11}\ \text{Pa}$

(b) Convert the area: $0.50\ \text{mm}^2 = 0.50 \times 10^{-6}\ \text{m}^2$ (1 mm² = 10⁻⁶ m²).

$$F = \sigma A = 240 \times 10^6 \times 0.50 \times 10^{-6} = 120 \text{ N}$$

$$\Delta x = \varepsilon x = 0.0020 \times 2.0 = 4.0 \times 10^{-3} \text{ m} = 4.0 \text{ mm}$$

Core Practical 3: the Young modulus of a wire

CORE PRACTICAL

Determine the Young modulus of a material

A long, thin wire (copper is typical) is clamped at one end of the bench and passes over a pulley at the other end, with a mass hanger on the end. A small marker (a piece of tape or a pointer) is fixed to the wire near the pulley, above a metre rule or vernier scale fixed to the bench.

1. Hang a small load to pull the wire straight and remove kinks. Then measure the **original length** x from the clamp to the marker with a metre rule or tape measure.
2. Measure the **diameter** with a **micrometer** at several places along the wire and in different directions; average them and calculate $A = \pi d^2/4$.
3. Add masses one at a time (for example 100 g each). After each, record the position of the marker; the extension Δx is the distance it has moved from its starting position.
4. Remove the masses one by one and check the marker returns to its start, which shows you stayed below the elastic limit.

Graph: calculate stress F/A and strain $\Delta x/x$ for each load and plot **stress against strain**. The gradient of the straight section is the Young modulus E. (Alternatively plot F against Δx: the gradient is EA/x, so $E = \text{gradient} \times x/A$.)

Why long and thin? $\Delta x = Fx/(EA)$, so a long, thin wire gives a larger extension, which can be measured with a smaller percentage uncertainty.

Sources of uncertainty and how to reduce them

- **Diameter**: the area depends on d^2, so its percentage uncertainty is doubled. This is usually the biggest uncertainty, which is why you use a micrometer and average several readings.
- **Extension**: it is small, so read the marker at eye level (avoid parallax), ideally with a vernier scale, and use a long wire.
- **Slipping** at the clamp or **friction** in the pulley give wrong extensions; make sure the clamp is tight and the pulley turns freely.
- Only use points on the straight part of the graph; points past the limit of proportionality curve away.

Safety: wear eye protection, because a wire that snaps can whip back. Put a box or sand tray under the masses and keep your feet clear.

WORKED EXAMPLE

In Core Practical 3 a copper wire has an original length of 2.50 m. Its diameter is measured as 0.38 ± 0.01 mm. A graph of load F against extension Δx is a straight line through the origin with a gradient of 5.2×10^3 N m^{-1}. (a) Calculate the Young modulus of copper. (b) Calculate the percentage uncertainty in the cross-sectional area.

Show solution

(a) The gradient is $F/\Delta x = EA/x$, so $E = \text{gradient} \times x/A$.

$$A = \frac{\pi d^2}{4} = \frac{\pi(0.38 \times 10^{-3})^2}{4} = 1.13 \times 10^{-7}\ \text{m}^2$$

$$E = \frac{5.2 \times 10^3 \times 2.50}{1.13 \times 10^{-7}} = 1.1 \times 10^{11}\ \text{Pa}$$

(b) Percentage uncertainty in $d = 0.01/0.38 \times 100 = 2.6\%$. Since $A \propto d^2$, the percentage uncertainty in A is $2 \times 2.6 = 5.3\%$. That is far bigger than the uncertainty in the 2.50 m length, which is why the diameter has to be measured so carefully.

Elastic strain energy

Stretching something takes work, and if the deformation is elastic that work is stored as **elastic strain energy**, ready to be released (think of a catapult or a bow). Work done is force × distance, but here the force isn't constant: it grows as the extension grows. The way round that is the area under the force–extension graph. Each thin strip under the graph is a small force × a small extra extension, a small piece of work, and adding all the strips gives the area.

KEY IDEA

The **area under a force–extension graph** is the work done in stretching the object. If the deformation is elastic, this is the elastic strain energy stored.

Elastic strain energy

FORMULA SHEET

$$E_{\text{el}} = \tfrac{1}{2} F \Delta x$$

E_{el}: elastic strain energy / J F: final force / N Δx: extension / m

Elastic strain energy from the area under the graph

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Below the limit of proportionality the force–extension graph is a straight line from the origin to the point $(\Delta x, F)$.

2. The area under it is a triangle with base Δx and height F:

$$E_{\text{el}} = \text{area} = \tfrac{1}{2} \times \text{base} \times \text{height}$$

$$\boxed{E_{\text{el}} = \tfrac{1}{2} F \Delta x}$$

Another way to see the ½: the force rises steadily from 0 to F, so the average force during the stretch is $F/2$, and work = average force × distance.

Elastic strain energy in terms of the stiffness

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Start from the area result:

$$E_{\text{el}} = \tfrac{1}{2} F \Delta x$$

2. Hooke's law gives $F = k\Delta x$. Substitute it in:

$$E_{\text{el}} = \tfrac{1}{2}(k\Delta x)\,\Delta x \;\Rightarrow\; \boxed{E_{\text{el}} = \tfrac{1}{2} k \Delta x^2}$$

Energy goes as the square of the extension: stretch a spring twice as far and it stores four times the energy.

COMMON MISTAKE

$\frac{1}{2}F\Delta x$ and $\frac{1}{2}k\Delta x^2$ only work while the graph is a **straight line**. For a curved graph you must find the area some other way, and if the object has been deformed **plastically**, not all the work you did is stored: the energy that went into moving atoms permanently is transferred to heating the material and is not returned when you let go.

Estimating the area under a curved graph

The spec expects you to estimate the energy for **non-linear** graphs too. The standard method is counting squares:

1. Work out what one grid square is worth: (force per square) × (extension per square) gives joules per square.
2. Count the whole squares under the curve, and count each part-square that is more than half under the curve as one (ignore those less than half).
3. Energy = number of squares × energy per square.

Alternatively split the area into simple shapes (triangles, rectangles, trapezia) and add their areas. Either way, smaller squares or thinner strips give a better estimate.

WORKED EXAMPLE

A catapult's elastic obeys Hooke's law. It is pulled back 0.15 m by a force that rises to 60 N, and it fires a 20 g stone. (a) Calculate the elastic strain energy stored. (b) Calculate the maximum possible speed of the stone. (c) Explain why the real speed will be lower.

Show solution

(a) $E_{\text{el}} = \frac{1}{2}F\Delta x = \frac{1}{2} \times 60 \times 0.15 = 4.5 \text{ J}$

(b) At most all of it becomes kinetic energy of the stone:

$$\tfrac{1}{2}mv^2 = 4.5 \;\Rightarrow\; v = \sqrt{\frac{2 \times 4.5}{0.020}} = 21 \text{ m s}^{-1}$$

(c) Some energy becomes kinetic energy of the elastic itself and some is dissipated as heat in the rubber, so the stone gets less than 4.5 J.

WORKED EXAMPLE

On a force–extension graph for a rubber band, each grid square is 0.5 N high and 1.0 cm wide. Up to an extension of 8.0 cm, a student counts 38 squares under the curve. Estimate the work done in stretching the band.

Show solution

One square represents $0.5 \text{ N} \times 0.010 \text{ m} = 5.0 \times 10^{-3} \text{ J}$.

$$W \approx 38 \times 5.0 \times 10^{-3} = 0.19 \text{ J}$$

Don't use $\frac{1}{2}F\Delta x$ here: the graph is curved, so the triangle would give the wrong area.

Quick checks

QUICK CHECK

Wires P and Q are made of the same metal. P is twice as long as Q and has twice the diameter. The same force is applied to each, within the limit of proportionality. What is (extension of P) / (extension of Q)?

- 4
- 1
- 0.5
- 0.25

$\Delta x = Fx/(AE)$. Doubling x doubles the extension, but doubling the diameter makes the area four times bigger, which divides it by 4. Overall $2/4 = 0.5$.

QUICK CHECK

Which statement describes the elastic limit?

- The point beyond which force is no longer proportional to extension.
- The largest force for which the object returns to its original length when the force is removed.
- The force at which the material breaks.
- The point where the material stretches with no increase in force.

The first option is the limit of proportionality, the third is (related to) the breaking stress and the fourth describes the yield point.

QUICK CHECK

A spring of stiffness 200 N m⁻¹ is stretched by 5.0 cm, within its limit of proportionality. How much elastic strain energy does it store?

- 0.25 J
- 0.50 J
- 5.0 J
- 250 J

$E_{\text{el}} = \frac{1}{2}k\Delta x^2 = \frac{1}{2} \times 200 \times 0.050^2 = 0.25$ J. 0.50 J forgets the ½; 250 J forgets to convert 5.0 cm into metres.

QUICK CHECK

Which of these is a correct unit for the Young modulus?

- N m⁻¹
- N m⁻²
- It has no unit, because strain has no unit.
- N m

Stress is in N m⁻² (Pa) and strain has no unit, so stress/strain is in N m⁻². N m⁻¹ is the unit of stiffness k, which belongs to an object, not a material.

QUICK CHECK

A wire is loaded beyond its elastic limit and then the load is removed. Which best describes the unloading part of its force–extension graph?

- It retraces the loading curve back to the origin.
- It is a straight line parallel to the initial straight part, ending at a positive extension.
- It drops vertically to zero force at the maximum extension.
- It is a straight line back to the origin.

The wire contracts with the same stiffness as it had in the elastic region, so the unloading line is parallel to the original straight line, but it doesn't get back to its original length: there is a permanent extension.

QUICK CHECK

A rubber band's force–extension graph curves so that it lies below the straight line joining the origin to its final point (Δx, F). The energy stored is

- exactly $\frac{1}{2}F\Delta x$
- exactly $F\Delta x$
- less than $\frac{1}{2}F\Delta x$, found from the area under the curve
- more than $\frac{1}{2}F\Delta x$, because rubber is elastic

The energy is always the area under the graph. If the curve sags below the straight line, the area under it is less than the triangle $\frac{1}{2}F\Delta x$. Estimate it by counting squares.